Is Overnight Volatility Overlooked?

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Abstract

The equity index futures, traded almost 24 hours, reveal the overnight price process. We use both daytime and overnight high-frequency futures price data to estimate the realized volatility (RV) of the S&P 500 and NASDAQ 100 indexes. We find that more than 40% of daily volatility is overnight volatility, which is omitted by many previous works. We demonstrate strong predictive power of overnight RV on daytime RV and vice versa. By incorporating overnight RV, our best proposed model reduces the forecasting mean squared error of the S&P 500 daytime RV by 27.8% compared to the benchmark model. Based on the inter-correlation between daytime and overnight volatility, we propose a novel Day-Night Realized Stochastic Volatility (DN-SV-RV) model, where the daytime and overnight returns are jointly modeled with their RVs, and their latent volatilities are correlated. The proposed DN-SV-RV model has the best out-of-sample return density forecasting among benchmark models. Under this innovative framework, the model estimation shows that daytime and overnight volatility complement each other, and volatility clustering is persistent during and after regular trading hours. Finally, by jointly estimating daily close-to-close return and realized volatility measures, we show that the daily RV, estimated from the almost 24-hour traded futures, is an accurate measure of daily return volatility by effectively measuring the overnight volatility, compared to daytime RV and squared overnight returns.

JEL Classification: C11 · C15 · C22 · C52

Keywords: Overnight volatility, Realized Volatility, Overnight Returns, Overnight Jumps, Stochastic Volatility, Bayesian MCMC, Density Prediction.

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"The stock market works by day, but it loves the night." — NY Times[1].

1 Introduction

The stock market is open during the business days, typically from 9:30 a.m. to 4:00 p.m. Eastern Standard Time. However, Cliff et al. (2008) report that the cumulative return from 1993 to 2006 by holding the SPY ETF is solely due to overnight returns. Sommer (2018) shows that all gains by holding the S&P 500 index (proxied by the SPY ETF) from 1993 to 2018 occur outside trading hours, and holding the equity-only portfolio during trading hours is an unprofitable investment strategy. Berkman et al. (2012) and Liu and Tse (2017) document this phenomenon for individual stocks, index ETFs and futures.

Information shocks are prone to arrive during the night. DeHaan et al. (2015) report that 21.5% of listed firms announced their earnings during trading hours in 2000, while only 2.2% firms did so in 2011. Important macroeconomic information announcements, e.g. the release of the unemployment rate, are usually scheduled at 8:30 a.m. EST before the opening of major exchanges (e.g., NYSE). Besides, global economic information, especially from Asia-Pacific and European countries, affects the U.S. equity market during the night. Unfortunately, few research, especially in field of financial time series modeling and forecasting, focuses on the overnight return and volatility series.

There are two fundamental issues for modeling overnight returns for major U.S. equity indexes. First, although the overnight return can be clearly defined as the price change during the non-trading hours, it would be misleading for major equity indexes like the S&P500 and Dow Jones Industrial Average (DJIA), since it depends how index provider sets the open price. The providers of the S&P500 and DJIA indexes have tended to set the open price according to the last close price for the last two decades. Ahoniemi and Lanne (2013) show that the overnight returns of the S&P 500 index are basically zero, especially before 2006, and they use the price of S&P 500 at 9:35 a.m. EST as the open price to calculate the overnight

[1] `https://www.nytimes.com/2018/02/02/your-money/stock-market-after-hours-trading.html`.

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returns of the S&P500 index. Moreover, global market indexes are also inconsistent in setting the open prices, as shown in Figure 5 in Appendix Section 7.1. Previously work that documents the importance of overnight returns, like Gallo (2001), Tsiakas (2008), Tseng et al. (2012), Ahoniemi and Lanne (2013), and Dhaene and Wu (2020), focus mainly on international equity indexes or individual stocks, which are free from the open price issue. In this paper, we use the open and close prices of index exchange-traded funds (ETFs) to evaluate the daytime and overnight returns. The price of ETFs can fluctuate during the pre-opening price discovery process, so that the overnight information is naturally embodied in the open price. Moreover, the Redemption Mechanism of the ETF excludes significant deviations of the ETF open price from the underlying constituents' open price.

Second, and more importantly, it is hard to measure the overnight volatility under the current market microstructure. The realized volatility, estimated from high-frequency daytime prices (Andersen and Bollerslev 1998; Barndorff-Nielsen and Shephard 2001), is a well-accepted consistent estimator of the volatility process. However, the absence of high-frequency overnight prices prevents researchers from directly measuring and studying the overnight volatility process. Many related papers use the squared overnight return (Fleming et al. 2003; Hansen and Lunde 2005; Ahoniemi and Lanne 2013; Todorova and Souček 2014; Fuertes et al. 2015; Maderitsch 2017) as a rough approximation to the overnight volatility. However, the squared overnight return, as a measure of overnight volatility, could be very noisy and hardly capture the dynamic of the overnight volatility process.

In this paper, we select the high-frequency price of overnight equity index futures (e.g. E-mini futures contracts) to estimate the overnight realized volatility. Taylor (2007) estimates the overnight volatility from S&P500 E-mini futures and provides its risk management (value-at-risk) implications. The continuous trading time of the futures market facilitates price discovery overnight. With the rapid development of the futures market since the early 2000s, overnight futures trading became increasingly important and contains more pertinent information. During the most recent COVID-19 pandemic, global stock markets became

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extremely volatile and erased $15 trillion in five weeks, since the end of February 2020. Within ten days in March 2020, the circuit breaker in the US stock market was triggered four times and the market suffered its third-largest daily percentage drop in history on 16th March 2020 (Zweig 2020; Zhang et al. 2020; Valetkevitch 2020). Moreover, even using the open price of major indexes, a great portion of daily stock crashes actually occur during the night. The future contracts, tied to those indexes, reflect the continuous process of price changes during the non-trading hours. In fact, even the futures contracts fell to their limit down thresholds (5% overnight) before the NYSE market opens.

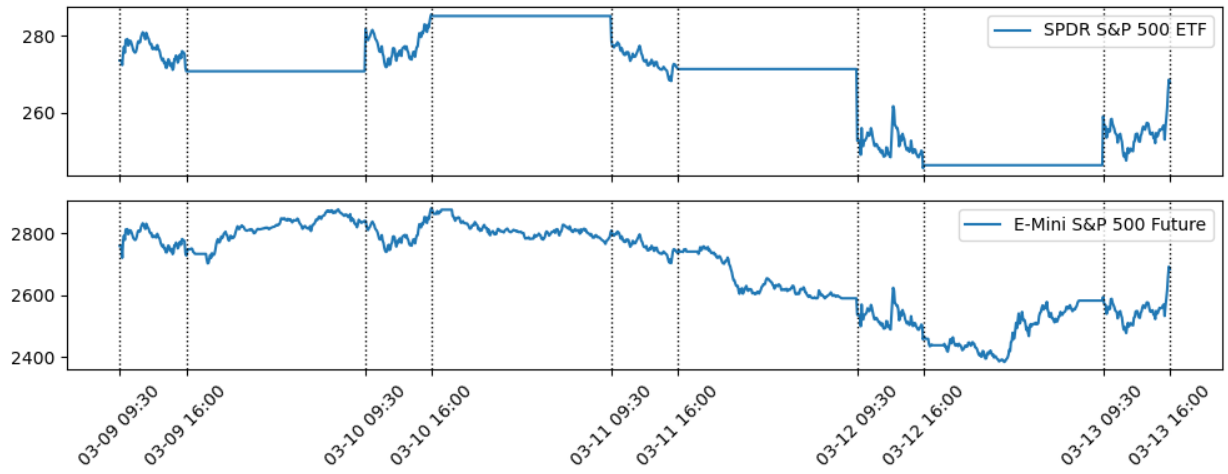


Figure 1: The price process of S&P 500 ETF and E-Mini future continuous contract between 2020-03-09 and 2020-03-13.

Figure 1 illustrates the prices of S&P 500 ETF (SPY) and S&P 500 E-Mini future continuous contract between 2020-03-09 and 2020-03-13. During the main trading hours, the ETF and the futures prices move in parallel, which is guaranteed by the market efficiency. However, during the non-trading hours, we can only observe a sudden price jump of the ETF at the market open. On the other hand, the futures price changes gradually during the non-trading hours. The observation here is not unique to the recent 2020 market crash. Within our sample period (2009-10-29 to 2020-07-01), the futures prices always reflect the process of overnight equity index changes. The overnight realized volatility estimated from the overnight futures prices contains incremental information which has been omitted by pre-

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vious works. Utilizing overnight information could substantially improve volatility modeling and forecasting.

Our paper makes the following contributions to the extant literature. First, we use the future prices to directly establish the estimation of the overnight realized volatility. Thus, our empirical estimation avoids the rough approximation when estimating the overnight effect on daytime realized volatility. Modeling and forecasting realized volatility is one of the major foci and challenges for financial econometricians (Andersen et al. 2003; Andersen et al. 2005; Gonçalves and Meddahi 2009; Corsi 2009; Patton 2011). The realized volatility measure has substantial asset pricing implications for underlying stock returns (Bollerslev et al. 2009b; Barndorff-Nielsen et al. 2008) and estimating stochastic volatility models (Deo et al. 2006; Takahashi et al. 2009). We follow the well-documented heterogeneous autoregression (HAR) process to re-examine the predictive capability of overnight realized volatility on the following daytime open to close realized volatility, which is the well-studied daily realized volatility of previous studies. For the S&P 500 (NASDAQ 100) index, including the overnight realized volatility will reduce the out-of-sample forecasting mean squared error of the following daytime logarithmic realized volatility by 16.9% (14.6%) compared to the benchmark HAR-RV model. Along with overnight return and implied volatility, our best model has a 27.8% (27.7%) lower out-of-sample forecasting mean squared error compared to the benchmark HAR-RV model for the S&P 500 (NASDA-100) index. The improvement is substantial, which is a significant contribution to the existing literature on realized volatility forecasting.

Moreover, we also model the overnight realized volatility with the HAR model and examine the predictive capability of daytime realized volatility for the following overnight close to open realized volatility. Similarly, we find significant improvement by incorporating daytime realized volatility. The estimation outcomes from the HAR models suggest strong correlation between daytime and overnight volatility.

Second, we propose a day-night realized stochastic volatility (DN-SV-RV) model to model

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jointly the daytime and overnight return and realized volatility. This innovative framework extends the SV model with realized volatility model (Takahashi et al. 2009). The model setup allows the inter-correlation between the daytime and overnight latent volatilities. Both the in-sample estimation and out-of-sample return distribution forecasts support a finding that the auto-correlation coefficient of the daytime (overnight) latent volatility will greatly decrease, which should be around 0.9 if modeled independently. We find that the daytime and overnight volatility processes are highly correlated, but not the same. The results suggest that the volatility clustering effect is persistent throughout the day and night. When predicting the daytime (overnight) volatility at market open (close), the most recent information is the immediately preceding overnight (daytime) volatility. Moreover, the mean of overnight returns, after controlling the stochastic volatility, is not necessarily higher than that of the daytime returns. The stochastic volatility sheds light on explaining the observed high overnight returns (Berkman et al. 2012; Branch and Ma 2012; Liu and Tse 2017).

Finally, we show that the daily close-to-close realized volatility, estimated from the almost 24-hour traded equity future prices, can be an accurate measure of the daily close-to-close return volatility. By jointly estimating the daily return and different realized volatility measure(s) (Takahashi et al. 2009), we show that the daily close-to-close realized volatility out-perform squared overnight returns in measuring the overnight volatility. Moreover, filling the overnight gap between the daytime realized volatility and the latent daily close-to-close return volatility, the daily close-to-close realized volatility could be useful measure for both academic or practical purposes.

The rest of the paper is organized as follows. Section 2 motivates the importance of incorporating overnight information by comparing the statistics of daytime and overnight returns and by estimating the daytime and overnight volatility process with the SV model. Section 2.3 explains the approach that we used to estimated the overnight realized volatility from high-frequency data from the futures market. We model and forecast daytime and overnight realized volatility, and confirm their strong correlation in Section 3. Section 4

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proposes the DN-SV-RV model and compares the model with benchmark SV and SV-RV models. Section 5 compares different daily realized volatility measures by jointly estimating daily return and volatility measure(s). Section 6 concludes the paper.

2 Motivations

2.1 Daily, Daytime and Overnight Returns

We use Eastern Standard Time and follow the trading calendar of the New York Stock Exchange (NYSE) throughout this paper. Given daily close price S_t^C, $t = 1, 2, ...T$, the daily close-to-close return is defined as:

$$r_t \equiv \log S_t^C - \log S_{t-1}^C.$$

This is the return used by most researchers and participants as the daily return. The daily return can be decomposed into two parts, the overnight close to open and the daytime open to close returns. We denote S_t^O the daily open price by $t = 1, 2, ...T$. The overnight return is defined as:

$$r_{N,t} \equiv \log S_t^O - \log S_{t-1}^C, \tag{1}$$

and the daytime return is then:

$$r_{D,t} \equiv \log S_t^C - \log S_t^O.$$

The daily return is the summation of the overnight return and daytime return ($r_t = r_{N,t} + r_{D,t}$). In this section, we demonstrate the behaviors of daytime and overnight returns of major equity indexes, index ETFs and individual stocks. We also discuss the issue of erroneous open prices of certain equity indexes. For the rest of this paper, all returns are scaled by 100 to represent percent returns.

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2.2 Significant Overnight Returns and Volatility

We illustrate how the open price of major equity indexes could be misleading, so that the overnight returns seem to be negligible. Figure 2 shows the daily returns and the overnight returns of three major equity indexes: the S&P 500, the DJIA and the NASDAQ 100 in the left column. The S&P 500 and DJIA overnight returns seem to be negligible before 2016, while the NASDAQ 100 overnight returns are significant throughout our sample period. Considering the fact that these indexes share many common constituents, the different overnight return behavior is quite puzzling. In Appendix Section 7.1, Figure 6 demonstrates the daily and overnight returns of sixteen representative companies included in the DJIA index. Given almost all constituents have significant overnight returns, the overnight return of indexes should also be significant.

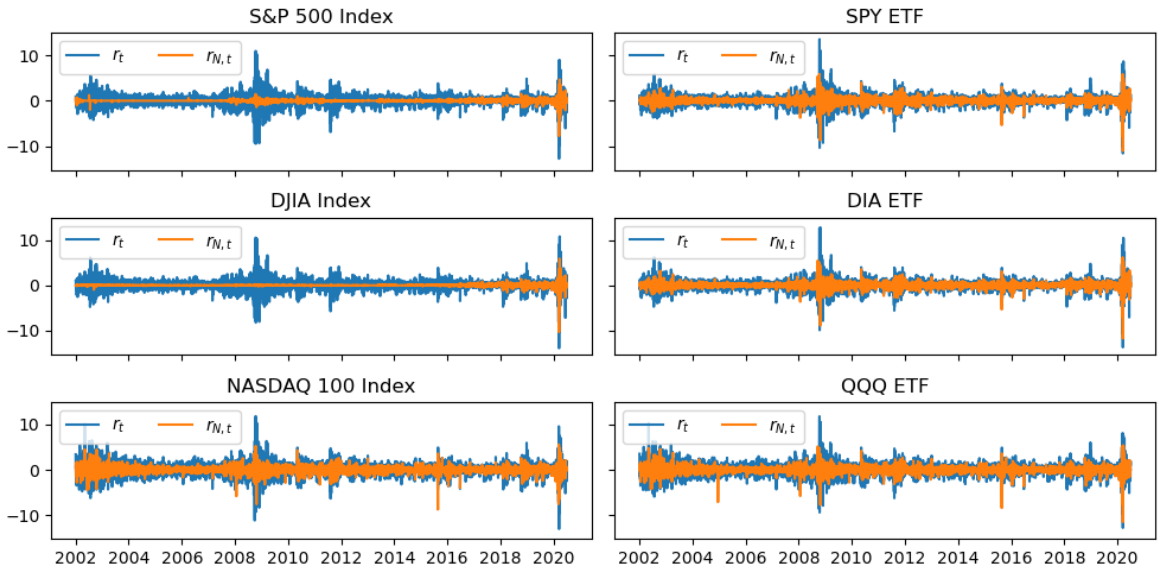


Figure 2: The daily and overnight return of major indexes and corresponding ETFs.

On the other hand, overnight returns are significant for all three ETFs as shown in the right column of Table 2. SPY and DIA overnight returns differ significantly from the underlying S&P 500 and DJIA indexes. The key difference between indexes and ETFs/individual stocks is that equity indexes are not directly tradable assets. For modern equity market, according to the NYSE, the opening auctions starts at 4:00 a.m. (Cao et al. 2000; Madhavan

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and Panchapagesan 2000; Madhavan and Panchapagesan 2002; Angel and Wu 2001), when limit orders can enter the market. From 4:00 a.m. to 9:30 a.m., the trading volume is little for most stocks. But major index ETFs, including their top holdings, have moving quotes due to the trading of index future contract. The pre-open price discovery reflects overnight information in the open price for individual stocks and ETFs. However, the index open price is not determined through this pre-opening price discovery process. As a result, the open prices of equity indexes do not necessarily represent the average opening prices of their component stocks, if the index provider casually sets the open price according to last close price.

For the rest of this paper, we use the 5-minute prices of the SPY and QQQ ETF, which are retrieved from Kibot database, to set the open and close prices. We use the price at market open (typically 9:30 a.m.) as the open price and the price at market close (typically 4:00 p.m.) as the close price[2]. Table 1 presents the summary statistics of daytime, overnight and daily returns. From Table 1, overnight returns have a higher mean and lower volatility, or higher Sharpe ratio, for the past ten years. The ratio of overnight volatility to daily volatility ($Var(r_{N,t})/Var(r_t)$) is 45% (40%) for the SPY (QQQ). More than 40% of daily volatility is overnight volatility. This observation challenges previous work, where the main focus is on modeling daytime volatility alone, by suggesting that 40% of the information in the volatility process is omitted.

	SPY			QQQ		
	r_t	$r_{N,t}$	$r_{D,t}$	r_t	$r_{N,t}$	$r_{D,t}$
Mean	0.0486	0.0268	0.0218	0.0711	0.0478	0.0233
Variance	1.1361	0.5104	0.5663	1.4369	0.5753	0.7967
Skewness	-0.7053	-2.5562	-0.5368	-0.5674	-2.1892	-0.4461
Kurtosis	14.0259	44.9277	5.9097	9.1864	34.2097	3.9199

Sample period: 2009-10-29 to 2020-07-01, totally 2686 observations.

Table 1: Summary statistics and correlation of daily, daytime and overnight returns.

In this paper, we model the overnight volatility process as a significant component of

[2]We also follow the NYSE calendar to set the early closure time.

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the daily close-to-close volatility. We concentrate on answering the question: Is overnight volatility process just a noise process? If not, is there volatility clustering for overnight volatility, and what's the relationship between daytime and overnight volatility? To answer these questions, we propose to measure the overnight volatility process as in the following sections.

2.3 Realized Volatility Measures and Jumps

We collect 5-minute high-frequency S&P 500 E-mini and NASDAQ 100 E-mini future prices from Kibot database. According to the CME Group, the trading hours of the E-mini contracts are Sunday 6:00 p.m. EST, to Friday 6:00 p.m. EST. After the close of trading on the NYSE, which is typically 4:00 p.m EST, there is a daily trading halt from 4:15 p.m. to 4:30 p.m. EST, Monday to Friday, and daily maintenance from 5:00 p.m. to 6:00 p.m. EST, Monday to Thursday. For every trading day, we match a continuous overnight trading interval starting at 6:00 p.m. EST on the previous calendar day to the following market open.

The log asset price $\log(S_t)$ follows a continuous jump-diffusion process:

$$d\log(S_t) = \mu_t dt + \sigma_t dW_t + J_t dq_t,$$

where W_t is the standard Brownian motion and dq_t is the Poisson process with jump intensity λ_t. J_t follows a normal distribution with mean μ_J and variance σ_J, measuring the jump size of the underlying process. Define the intraday return over time interval Δ as:

$$r_{t,j} = \log(S_{t,j\cdot\Delta}) - \log(S_{t,(j-1)\cdot\Delta}),$$

where $\log(S_{t,j\cdot\Delta})$ is the log asset price at day t and time $j\cdot\Delta$. As shown in Barndorff-Nielsen

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and Shephard (2004), the realized variance and bipower variation of asset returns are

$$RV_t = \sum_{j=1}^{m} r_{t,j}^2 \xrightarrow{\Delta \downarrow 0} \int_{t-1}^{t} \sigma_s^2 ds + \int_{t-1}^{t} J_s^2 dq_s, \tag{2}$$

$$BV_t = \frac{\pi}{2} \frac{m}{m-1} \sum_{j=1}^{m} |r_{t,j}||r_{t,j-1}| \xrightarrow{\Delta \downarrow 0} \int_{t-1}^{t} \sigma_s^2 ds. \tag{3}$$

We also filter the cumulative squared jumps (Barndorff-Nielsen and Shephard 2004; Huang and Tauchen 2005; Andersen et al. 2007; Tauchen and Zhou 2011) as:

$$J_t^2 = (RV_t - BV_t) \times I_{(ZJ_t \geq \Phi_\alpha^{-1})}. \tag{4}$$

Where

$$ZJ_t = \frac{RV_t - BV_t}{RV_t \sqrt{\left[\left(\frac{\pi}{2}\right)^2 + \ - 5\right] \frac{1}{m} \max\left(1, \frac{TP_t}{BV_t^2}\right)}} \xrightarrow{d} N(0,1), \tag{5}$$

which converges to a standard normal distribution and

$$TP_t = m\mu_{4/3}^{-3} \frac{m}{m-2} \sum_{j=3}^{m} |r_{t,j-2}|^{4/3} |r_{t,j-1}|^{4/3} |r_{t,j}|^{4/3} \to \int_{t-1}^{t} \sigma_s^4 ds, \tag{6}$$

where $\mu_k = 2^{k/2}\Gamma((k+1)/2)/\Gamma(1/2)$ for $k > 0$. The squared cumulative jump is then filtered at significance level α of the z-test. For this paper, we choose $\alpha = 0.99$.

2.4 Overnight Realized Volatility and Jumps

Figure 3 illustrates how we estimate the daytime, overnight and daily realized volatility and set the time label. We use the 5-minute prices to calculate the intraday returns. Using returns within the overnight time interval (from 6:00 p.m. of the previous calendar day to 9:30 a.m. of day t), we estimate the overnight realized volatility, denoted as $RV_{N,t}$. Similarly, daytime realized volatility $RV_{D,t}$ is estimated with intraday returns within trading hours. In addition, we can also estimate the daily realized volatility RV_t with intraday returns from

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6:00 p.m. of the previous calendar day $t-1$ to 4:00 p.m. (1:00 p.m. if early closure) of day t. Given our notation, $RV_{N,t}$, although labeled with day t, is observed before $RV_{D,t}$ and RV_t=$RV_{N,t} + RV_{D,t}$.

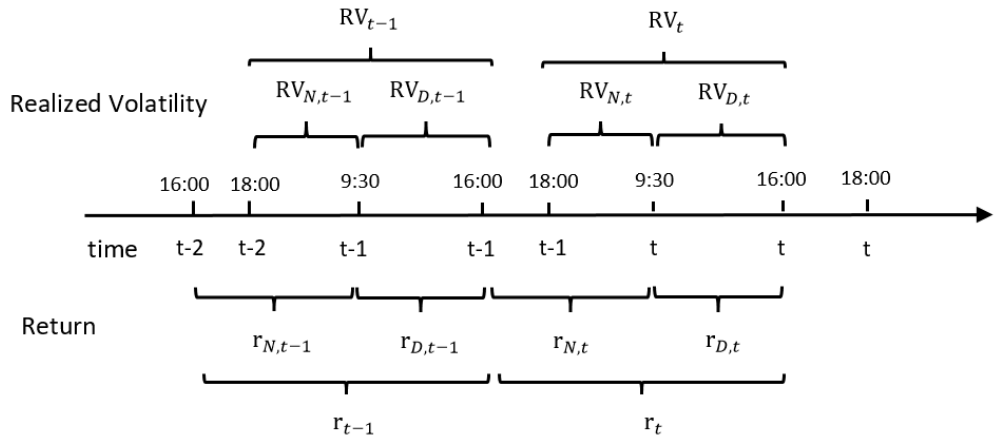


Figure 3: The time label of the daily, daytime and overnight RV and returns.

Notice that during the trading hours, both the ETF and E-Mini futures are traded. The daytime realized volatility measures calculated from the ETF and futures are almost identical. Considering the higher trading volume of the futures market and the consistency with the overnight realized volatility, we will use the daytime realized volatility calculated from futures prices in the following. In total, we collect 2,560 daily observations from 2009-08-29 to 2019-12-31.

Similarly, we can also estimate the overnight, daytime and daily bi-power variation ($BV_{N,t}$, $BV_{D,t}$ and BV_t) and squared overnight jumps ($J^2_{N,t}$, $J^2_{D,t}$ and J^2_t). We report the summary statistics of the realized volatility measures of the S&P 500 in Table 2. The summary statistics table for the NASDAQ 100 can be found in the Appendix, Table 8.

The means of realized volatility and bi-power variation (daily, daytime and overnight respectively) match the variances of the corresponding returns in Table 1, which supports the accuracy of our realized volatility measures. The correlation coefficient between the overnight and daytime realized volatility, which is a lead-lag correlation since $RV_{N,t}$ is observed before $RV_{D,t}$, is 0.7583. This confirms the high positive correlation between overnight and daytime

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	RV_t	$RV_{N,t}$	$RV_{D,t}$	BV_t	$BV_{N,t}$	$BV_{D,t}$
Mean	1.1576	0.4872	0.6704	1.0944	0.4327	0.6447
Var	10.9895	2.2700	4.0952	10.7188	2.0373	3.8680
RV_t	1.0000					
$RV_{N,t}$	0.9174	1.0000				
$RV_{D,t}$	0.9551	0.7583	1.0000			
BV_t	0.9965	0.9201	0.9474	1.0000		
$BV_{N,t}$	0.8775	0.9888	0.7013	0.8876	1.0000	
$BV_{D,t}$	0.9639	0.7870	0.9930	0.9609	0.7347	1.0000

Sample period: 2009-10-29 to 2020-07-01, totally 2686 observations.

Table 2: Summary statistics and correlations of S&P 500 realized volatility measures.

volatility, but also challenges the method of estimating the whole day realized volatility by scaling the daytime realized volatility upward to match the daily close-to-close return volatility. Scaling assumes that the daytime and overnight realized volatility are perfectly correlated, which is not true given our results. The overnight volatility contains information that is independent of the daytime volatility and should not be omitted.

The left column of Figure 4, from top to bottom, demonstrates the daytime, overnight and daily realized volatility ($RV_{D,t}$, $RV_{N,t}$ and RV_t respectively) of SPY. Given our estimation of realized volatility measures, the daily realized volatility is the summation of daytime and overnight realized volatility (RV_t=$RV_{N,t}$+$RV_{D,t}$). The fourth row shows the squared overnight returns ($r^2_{N,t}$). The right column is the logarithmic of the left column. Comparing $\log RV_{D,t}$ and $\log RV_{N,t}$, it is obvious that the overnight volatility is not a noise process. $\log RV_{N,t}$ series shares a similar stochastic volatility pattern with the $\log RV_{N,t}$ series. The well-documented heteroscedasticity and volatility clustering also apply to overnight volatility. During the pandemic crisis (March 2020), we observe significantly higher $\log RV_{D,t}$, $\log RV_{N,t}$ and $\log RV_t$. On the other hand, $\log r^2_{N,t}$ series could be too noisy, as an overnight volatility measure, and a great portion of overnight information could be missed by modeling with $r^2_{N,t}$ (or $\log r^2_{N,t}$).

The last row of Figure 4 shows the filtered daytime (left) and overnight (right) squared jumps. They share the same vertical axis for comparison purposes. It is obvious that jumps

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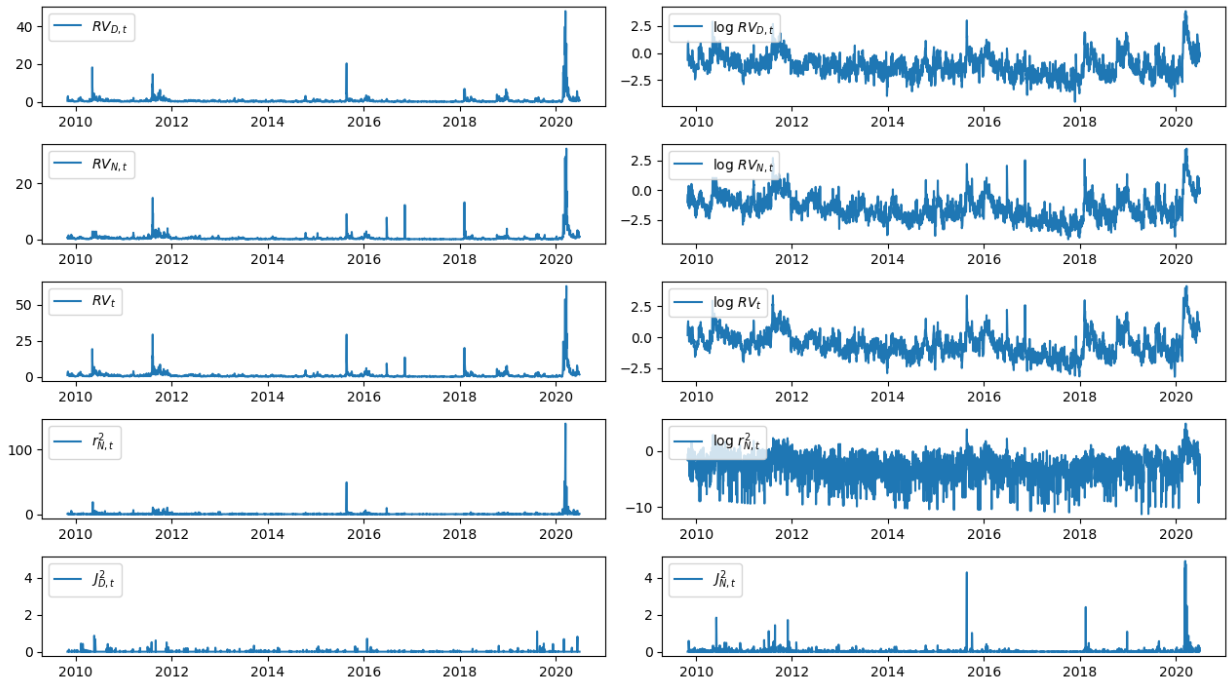


Figure 4: SPY realized volatility measures and squared jumps.

are prone to occur outside the main trading hours. This is consistent with the observation in Table 1 that the kurtosis of SPY (QQQ) overnight returns is 44.93 (34.21), which is significantly higher than that of the daytime returns. Releasing important information outside the main trading hours, instead of avoiding shocks in the equity market, merely postpones the shocks. The estimated overnight squared jumps reflect the shocks from the overnight-traded futures market, especially during the pandemic crisis. We observe significant overnight realized jumps in March 2020 in Figure 4.

Using the high-frequency futures prices, we reveal the overnight volatility and jumps by estimating the realized volatility measures and squared jumps. There will be numerous economic implications, and the scope may not limit to what we concentrate in this paper. We focus on the daytime and overnight volatility. In the next section, we model the daytime and overnight realized volatility process and investigate the correlation between daytime and overnight volatility.

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3 Forecasting Daytime and Overnight RV

3.1 HAR-RV modeling of Realized Volatility

Predicting the future realized volatility, especially after assuming a constant return mean, is the key component in return distribution prediction (Bollerslev et al. 2009a; Maheu and McCurdy 2011). The heterogeneous autoregressive model for realized volatility (HAR-RV), developed by Corsi (2009), is popular for modeling the realized volatility process since it captures the long memory of the volatility process with a simple specification. Inasmuch as Andersen et al. (2003) have shown that the distribution of logarithmic realized volatility is close to normal, we will model the logarithmic realized volatility like Andersen et al. (2007) and Liu and Maheu (2009) and follow a parameterization similar to Patton and Sheppard (2015) for clear parameter interpretation.

First, we model the daytime realized volatility and focus on forecasting the following daytime realized volatility given all the information available at the market open. The basic HAR-RV model is specified as follows:

$$\log RV_{D,t} = \mu + \rho_1 \log RV_{D,t-1} + \rho_5 \left(\frac{1}{4} \sum_{i=2}^{5} \log RV_{D,t-i} \right) + \rho_{22} \left(\frac{1}{17} \sum_{i=6}^{22} \log RV_{D,t-i} \right) + e_t. \tag{7}$$

Recent work (Corsi and Renò 2012; Wang et al. 2015) extend the HAR-RV by including the implied volatility, leverage effect, etc. Before adding more explanatory variables to capture more volatility-related effects or modifying the HAR-RV model with non-linear, non-Gaussian properties like Corsi et al. (2008), we need to ensure that the right hand side includes all the available information of the volatility process itself at the market open.

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Including the overnight realized volatility leads to Eq. (8):

$$\begin{aligned}\log RV_{D,t} =&\mu + \rho_1 \log RV_{D,t-1} + \rho_5 \left(\frac{1}{4}\sum_{i=2}^{5} \log RV_{D,t-i}\right) + \rho_{22} \left(\frac{1}{17}\sum_{i=6}^{22} \log RV_{D,t-i}\right) \\ &+ \theta_1 \log RV_{N,t} + \theta_5 \left(\frac{1}{4}\sum_{i=1}^{4} \log RV_{N,t-i}\right) + \theta_{22} \left(\frac{1}{17}\sum_{i=5}^{21} \log RV_{N,t-i}\right) + e_t. \quad (8)\end{aligned}$$

We include positive and negative overnight returns $r^+_{N,t} = \max(r_{N,t}, 0)$, $r^-_{N,t} = \min(r_{N,t}, 0)$, and lagged daytime returns $r^+_{D,t} = \max(r_{D,t}, 0)$, $r^-_{D,t} = \min(r_{D,t}, 0)$ to control the asymmetry between return and volatility (Corsi and Renò 2012). We also include the lagged logarithmic implied volatility ($\log IV_{t-1}$, $\frac{1}{4}\sum_{i=2}^{5} \log IV_{t-i}$ and $\frac{1}{17}\sum_{i=6}^{22} \log IV_{t-i}$). For SPY (QQQ), the logarithmic implied volatility is calculatd from the VIX (VXN) index ($\log IV_t = 2(\log VIX_t - \log 21)$). Notice IV_{t-1} denotes the closing price of the volatility index so that $\log IV_{t-1}$ does not contain overnight information. We can classify all the right hand side variables into two sets: variables with and without overnight information. We expect that variables with overnight information (overnight realized volatility and overnight returns) should have strong predictive power.

To predict the overnight realized volatility at the market close, the benchmark HAR model is given by Eq. (9):

$$\log RV_{N,t} = \mu + \rho_1 \log RV_{N,t-1} + \rho_5 \left(\frac{1}{4}\sum_{i=2}^{5} \log RV_{N,t-i}\right) + \rho_{22} \left(\frac{1}{17}\sum_{i=6}^{22} \log RV_{N,t-i}\right) + e_t. \quad (9)$$

And we extend Eq. (9) with lagged daytime realized volatility as follows:

$$\begin{aligned}\log RV_{N,t} =&\mu + \rho_1 \log RV_{N,t-1} + \rho_5 \left(\frac{1}{4}\sum_{i=2}^{5} \log RV_{N,t-i}\right) + \rho_{22} \left(\frac{1}{17}\sum_{i=6}^{22} \log RV_{N,t-i}\right) \\ &\theta_1 \log RV_{D,t-1} + \theta_5 \left(\frac{1}{4}\sum_{i=2}^{5} \log RV_{D,t-i}\right) + \theta_{22} \left(\frac{1}{17}\sum_{i=6}^{22} \log RV_{D,t-i}\right) + e_t. \quad (10)\end{aligned}$$

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Similarly, we also include positive and negative lagged daytime returns ($r^{+}_{D,t-1}$, $r^{-}_{D,t-1}$), lagged overnight returns ($r^{+}_{N,t-1}$, $r^{-}_{N,t-1}$) and implied volatility ($\log IV^{O}_{t-1}$, $\frac{1}{4}\sum_{i=2}^{5}\log IV^{O}_{t-i}$ and $\frac{1}{17}\sum_{i=6}^{22}\log IV^{O}_{t-i}$). We choose the opening price of the volatility index so that $\log IV^{O}_{t-1}$ does not contain the lagged daytime information when predicting the overnight realized volatility at the market close. We can also classify the right hand variables for overnight realized volatility forecasting into two sets: variables with and without lagged daytime information.

We estimated the HAR-RV model using the Bayesian MCMC method. For model comparison, we calculate the forecasting mean squared error ($FMSE$) to measure the accuracy of mean forecasting and the logarithmic predictive likelihood (LPL) to measure the accuracy of distributional forecasting (Geweke and Amisano 2010; Gelman et al. 2014). FMSE and LPL are defined as follows:

$$FMSE_M = \frac{1}{T-s}\sum_{t=s}^{T-1}\left(\left(y_{t+1}-\hat{y}_{M,t+1}\right)^2|\mathcal{F}_t\right), \tag{11}$$

$$LPL_M = \sum_{t=s}^{T-1}\log p_M(y_{t+1}|\mathcal{F}_t). \tag{12}$$

Here y_{t+1} is the observation to predict and $\hat{y}_{M,t+1}$ is the prediction of y_{t+1} with available information at time t ($\mathcal{F}_t$ denotes the available information set) according to model M. Since the HAR model is essentially a linear model with normally distributed noise, $p_M(y_{t+1}|\mathcal{F}_t)$ is a normal likelihood. Let $x_{1:t}$ denote the set of explanatory variable vectors $x_1, x_2, ...x_t$ and $y_{1:t}$ denote the dependent variable vector $y_1, y_2, ...y_t$. By fitting the HAR model M with $y_{1:t}$ and $x_{1:t}$, we have the MCMC draws of correlation coefficients $\beta^{(i)}_{M,t}$ and variance $\sigma^{(i)}_{M,t}$ ($i = 1, 2...N$, N denotes the number of MCMC iterations). Then $\hat{y}_{M,t+1}$ and $p_M(y_{t+1}|\mathcal{F}_t)$ of model M can be estimated as:

$$\hat{y}_{M,t+1} \approx \frac{1}{N}\sum_{i=1}^{N} {x_{t+1}}'\beta^{(i)}_{M,t}, \tag{13}$$

$$p_M(y_{t+1}|\mathcal{F}_t) \approx \frac{1}{N}\sum_{i=1}^{N} f_N(y_{t+1}|{x_{t+1}}'\beta^{(i)}_{M,t}, \sigma^{(i)}_{M,t}). \tag{14}$$

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$f_N(x|\mu, \sigma^2)$ denotes the likelihood function of a normal distribution with mean μ and variance σ^2. The total sample size is 2686 ($T = 2686$). We set the initial sample size to be 86 (s=86) and the number of MCMC iteration to be 10,000 ($N = 10,000$). We also set 1,000 burn-in iterations for parameter convergence. Clearly, a lower $FMSE_M$ and a higher LPL_M indicate better predictive performance of model M. To compare two models M_1 and M_2, we focus on the logarithmic predictive Bayes factor (LBF), which is defined as: $LPL_{M_1} - LPL_{M_2}$. A logarithmic predictive Bayes factor greater than 5 lends strong support for model M_1 over model M_2. In addition, we also compare two models by the percentage change of $FMSE$ ($FMSE$ $P.C.$ $= (FMSE_{M_1} - FMSE_{M_2})/FMSE_{M_2}$).

Tables 3 summarizes the empirical results of forecasting SPY daytime realized volatility with HAR models. The benchmark model is column 1 in Table 3. The last fours rows of Tables 3 report the forecasting mean squared errors ($FMSE$) and logarithmic predictive likelihood (LPL) of each model, including the percentage change of $FMSE$ ($FMSE$ $P.C.$) and logarithmic predictive Bayes factor (LBF) compared to the benchmark HAR-RV model.

As shown by column 2, Table 3, Including $r^+_{D,t}$ and $r^-_{D,t}$ leads to a LBF of 58.73 compared to the benchmark HAR-RV model. Further including the implied volatility, as shown by column 3 of Table 3, significantly improves the volatility forecasting as the LBF ($FMSE$ $P.C.$) is 187.44 (-13.2%), which is a strong evidence of forecasting improvement. The results confirm that the asymmetric return-volatility correlation and the implied volatility, as an *ex-ante* volatility measure, are important for volatility forecasting.

Columns 5 (4) of Table 3 shows that including the overnight realized volatility (positive and negative overnight returns) leads to a logarithmic Bayes factor of 242.32 (131.19), which is strongly significant. Including both overnight realized volatility and overnight returns (column 6 of Table 3) produces a logarithmic Bayes factor as high as 319.67. According to the $FMSE$, incorporating overnight realized volatility alone will decrease the $FMSE$ by 16.9%, which is greater than the best model without overnight information (as shown by column 3 of Table 3). Including both overnight realized volatility and return will decrease

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the $FMSE$ by 21.67%. The results strongly support the value of overnight information in forecasting the daytime realized volatility. As shown by column 7 of Table 3, our best model decreases the $FMSE$ by 27.81% compared to the benchmark HAR-RV model. The improvement is substantial and contributes to the current researches on realized volatility forecasting.

Regarding the parameters, overnight realized volatility $\log RV_{N,t}$ has a positive effect on the following daytime realized volatility $\log RV_{D,t}$, which suggests that higher overnight volatility is likely to be followed by higher daytime volatility. Consistent with the well-documented leverage effect, overnight returns are negatively correlated with the following daytime realized volatility. Moreover, if we compare daytime and overnight returns, as shown by column 2 and 4 of Table 3, the $r^+_{D,t}$ has weaker positive effect while $r^-_{D,t}$ has stronger negative effect on the next daytime volatility. Corsi and Renò (2012) use only negative daily returns to control the asymmetry. For overnight returns, we find that both $r^+_{N,t}$ and $r^-_{N,t}$ has negative effect on the following daytime volatility. The empirical results suggest that if the equity index increases (decreases) during the night, following daytime market tends to be less (more) volatile.

We are not searching for the best model (highest logarithmic predictive likelihood) for realized volatility forecasting in this paper. We present several HAR models to confirm that the overnight information, the overnight realized volatility, and overnight returns, are important for volatility modeling and forecasting. If we are forecasting the following daytime volatility at market open, the most recent market information available is contained in the overnight tradings. The overnight realized volatility, estimated from high-frequency index future prices, and the overnight returns, calculated from index ETF close and open prices[3], should be included for further extend models of volatility forecasting.

Tables 4 summarizes the results of forecasting SPY overnight realized volatility. The

[3]As we have discussed, given the open prices of major index fail to fully capture the overnight price changes, overnight returns calculated from the index close and open prices lead to poor forecasting performance.

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benchmark model is column 1 in Table 4. As expected, the lagged daytime realized volatility yields the greatest marginal improvement (column 5 of Table 4 with a logarithmic Bayes factor of 334.59 and 22.64% decrease in $FMSE$), followed by the lagged positive and negative daytime returns (column 4 of Table 4 with a logarithmic Bayes factor of 204.45 and 14.47% percent decrease in $FMSE$). The lagged daytime realized volatility $\log RV_{D,t-1}$ has a positive effect on the following overnight realized volatility $\log RV_{N,t}$. When forecasting the overnight volatility, the most recent information available is contained in the daytime tradings and we expect that the daytime realized volatility and returns have strong forecasting power. In addition, we find significant predictive power of implied volatility (the open price of the implied volatility) on the overnight realized volatility, which also supports the informativeness of implied volatility.

The inter-correlations between daytime and overnight realized volatility indicate that the volatility process is continuous and higher overnight volatility is likely to be followed by higher daytime volatility and vice versa. Consistent with the leverage effect, overnight return (lagged daytime return) is negatively correlated with the following daytime (overnight) realized volatility. All findings for the realized volatility of the S&P 500 index in this section are consistent with those for the NASDAQ 100 index. Tables 9 and 10 in the Appendix Section 7.3 summarize the results of the HAR-RV models for the NASDAQ 100 index. Consistent results support the robustness of our conclusion.

The strong correlation between the daytime and overnight realized volatility calls for jointly modeling the daytime and overnight return and realized volatility, which captures the inter-correlations between the daytime and overnight volatility processes. A potential concern against modeling the realized volatility itself, like HAR-RV models, could be that the estimated realized volatility, especially the overnight realized volatility, may not be an accurate measure of the real return volatility. In the next section, we will propose a day-night stochastic volatility model with realized volatility, in which both daytime and realized volatility measures will be treated as measures of the real volatility process with noises.

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	(1)	(2)	(3)	(4)	(5)	(6)	(7)
μ	-0.0858	-0.2569	-0.3030	-0.1311	-0.0160	-0.0025	-0.2066
	[0.019]	[0.028]	[0.032]	[0.025]	[0.023]	[0.028]	[0.046]
	(-0.124, -0.048)	(-0.313, -0.202)	(-0.365, -0.241)	(-0.180, -0.082)	(-0.060, 0.030)	(-0.057, 0.052)	(-0.296, -0.116)
$\log RV_{D,t-1}$	0.5791	0.4777	0.2705	0.5700	0.3217	0.3536	0.1902
	[0.018]	[0.020]	[0.022]	[0.017]	[0.020]	[0.019]	[0.021]
	(0.543, 0.614)	(0.438, 0.516)	(0.227, 0.314)	(0.536, 0.603)	(0.283, 0.361)	(0.316, 0.392)	(0.148, 0.232)
$\frac{1}{4}\sum_{i=2}^{5}\log RV_{D,t-i}$	0.2688	0.2940	0.2670	0.2504	0.2201	0.2046	0.2477
	[0.023]	[0.023]	[0.033]	[0.022]	[0.034]	[0.033]	[0.035]
	(0.224, 0.313)	(0.248, 0.339)	(0.203, 0.331)	(0.207, 0.294)	(0.154, 0.286)	(0.140, 0.268)	(0.180, 0.317)
$\frac{1}{17}\sum_{i=6}^{22}\log RV_{D,t-i}$	0.0780	0.0930	0.1553	0.0879	0.1956	0.1940	0.2591
	[0.020]	[0.019]	[0.038]	[0.019]	[0.048]	[0.046]	[0.046]
	(0.039, 0.116)	(0.055, 0.131)	(0.081, 0.231)	(0.052, 0.125)	(0.103, 0.290)	(0.102, 0.284)	(0.167, 0.350)
$r^{+}_{D,t-1}$	-	0.0889	0.1246	-	-	-	0.1038
		[0.029]	[0.028]				[0.026]
		(0.031, 0.148)	(0.069, 0.178)				(0.052, 0.154)
$r^{-}_{D,t-1}$	-	-0.3141	-0.1616	-	-	-	-0.1326
		[0.028]	[0.028]				[0.026]
		(-0.370, -0.259)	(-0.217, -0.105)				(-0.183, -0.081)
$r^{+}_{N,t}$	-	-	-	-0.1614	-	-0.1974	-0.2494
				[0.030]		[0.028]	[0.027]
				(-0.220, -0.103)		(-0.251, -0.143)	(-0.302, -0.196)
$r^{-}_{N,t}$	-	-	-	-0.3087	-	-0.1730	-0.1665
				[0.023]		[0.023]	[0.022]
				(-0.355, -0.263)		(-0.218, -0.128)	(-0.209, -0.124)
$\log RV_{N,t}$	-	-	-	-	0.4979	0.4391	0.3182
					[0.021]	[0.022]	[0.022]
					(0.456, 0.539)	(0.397, 0.481)	(0.275, 0.361)
$\frac{1}{4}\sum_{i=1}^{4}\log RV_{N,t-i}$	-	-	-	-	-0.1275	-0.0896	-0.0950
					[0.038]	[0.037]	[0.038]
					(-0.202, -0.052)	(-0.161, -0.018)	(-0.169, -0.020)
$\frac{1}{17}\sum_{i=5}^{21}\log RV_{N,t-i}$	-	-	-	-	-0.1782	-0.1664	-0.1746
					[0.049]	[0.047]	[0.054]
					(-0.275, -0.084)	(-0.257, -0.076)	(-0.281, -0.070)
$\log IV_{t-1}$	-	-	1.1744	-	-	-	0.8852
			[0.073]				[0.070]
			(1.030, 1.317)				(0.749, 1.024)
$\frac{1}{4}\sum_{i=2}^{5}\log IV_{t-i}$	-	-	-0.6821	-	-	-	-0.5238
			[0.090]				[0.085]
			(-0.855, -0.502)				(-0.690, -0.357)
$\frac{1}{17}\sum_{i=6}^{22}\log IV_{t-i}$	-	-	-0.2222	-	-	-	-0.1274
			[0.066]				[0.069]
			(-0.350, -0.094)				(-0.262, 0.006)
FMSE	0.3390	0.3242	0.2943	0.3074	0.2817	0.2656	0.2447
FMSE P.C.	-	-4.37%	-13.20%	-9.32%	-16.90%	-21.67%	-27.81%
LPL	-2287.95	-2229.22	-2100.51	-2156.76	-2045.63	-1968.28	-1860.79
LBF	-	58.73	187.44	131.19	242.32	319.67	427.16

The table reports the posterior mean, standard deviation in [] and 0.95 density interval in ().

Table 3: Forecasting S&P500 daytime logarithmic realized volatility with HAR models

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	(1)	(2)	(3)	(4)	(5)	(6)	(7)
μ	-0.0888	-0.1575	-0.3625	-0.2572	-0.0904	-0.1480	-0.3736
	[0.021]	[0.028]	[0.045]	[0.027]	[0.021]	[0.027]	[0.043]
	(-0.130, -0.047)	(-0.212, -0.103)	(-0.451, -0.274)	(-0.310, -0.203)	(-0.131, -0.050)	(-0.199, -0.095)	(-0.459, -0.288)
$\log RV_{N,t-1}$	0.5193	0.4676	0.2549	0.4666	0.1967	0.2341	0.1323
	[0.018]	[0.019]	[0.023]	[0.018]	[0.020]	[0.020]	[0.021]
	(0.483, 0.556)	(0.430, 0.506)	(0.210, 0.299)	(0.432, 0.502)	(0.158, 0.236)	(0.195, 0.273)	(0.090, 0.173)
$\frac{1}{4}\sum_{i=2}^{5}\log RV_{N,t-i}$	0.3515	0.3696	0.2909	0.3418	0.2760	0.2660	0.2738
	[0.024]	[0.024]	[0.034]	[0.022]	[0.031]	[0.031]	[0.035]
	(0.304, 0.398)	(0.323, 0.416)	(0.226, 0.358)	(0.299, 0.385)	(0.214, 0.337)	(0.205, 0.325)	(0.206, 0.342)
$\frac{1}{17}\sum_{i=6}^{22}\log RV_{N,t-i}$	0.0693	0.0788	0.1058	0.0856	0.2971	0.2877	0.2074
	[0.020]	[0.019]	[0.044]	[0.018]	[0.041]	[0.040]	[0.051]
	(0.031, 0.108)	(0.042, 0.117)	(0.019, 0.192)	(0.050, 0.121)	(0.217, 0.380)	(0.211, 0.366)	(0.107, 0.305)
$r^{+}_{N,t-1}$	-	-0.0409	-0.0246	-	-	-	0.0387
		[0.029]	[0.027]				[0.025]
		(-0.098, 0.017)	(-0.078, 0.030)				(-0.010, 0.087)
$r^{-}_{N,t-1}$	-	-0.2088	-0.1412	-	-	-	-0.1256
		[0.024]	[0.023]				[0.021]
		(-0.255, -0.162)	(-0.186, -0.096)				(-0.166, -0.084)
$r^{+}_{D,t-1}$	-	-	-	-0.0193	-	-0.0674	-0.0915
				[0.027]		[0.024]	[0.024]
				(-0.071, 0.033)		(-0.116, -0.020)	(-0.139, -0.044)
$r^{-}_{D,t-1}$	-	-	-	-0.4259	-	-0.2204	-0.2306
				[0.023]		[0.023]	[0.023]
				(-0.471, -0.381)		(-0.266, -0.175)	(-0.275, -0.187)
$\log RV_{D,t-1}$	-	-	-	-	0.4635	0.3738	0.2937
					[0.017]	[0.019]	[0.019]
					(0.430, 0.497)	(0.337, 0.410)	(0.256, 0.332)
$\frac{1}{4}\sum_{i=2}^{5}\log RV_{D,t-i}$	-	-	-	-	-0.0057	0.0242	-0.0129
					[0.031]	[0.031]	[0.031]
					(-0.067, 0.056)	(-0.035, 0.084)	(-0.074, 0.048)
$\frac{1}{17}\sum_{i=6}^{22}\log RV_{D,t-i}$	-	-	-	-	-0.2385	-0.2206	-0.1637
					[0.042]	[0.040]	[0.040]
					(-0.322, -0.157)	(-0.299, -0.142)	(-0.243, -0.085)
$\log IV^{O}_{t-1}$	-	-	0.9586	-	-	-	0.5687
			[0.065]				[0.061]
			(0.832, 1.085)				(0.449, 0.689)
$\frac{1}{4}\sum_{i=2}^{5}\log IV^{O}_{t-i}$	-	-	-0.3758	-	-	-	-0.2374
			[0.083]				[0.076]
			(-0.537, -0.213)				(-0.385, -0.091)
$\frac{1}{17}\sum_{i=6}^{22}\log IV^{O}_{t-i}$	-	-	-0.1807	-	-	-	-0.0317
			[0.070]				[0.064]
			(-0.319, -0.044)				(-0.158, 0.093)
FMSE	0.2991	0.2888	0.2632	0.2558	0.2314	0.2196	0.2083
FMSE P.C.	-	-3.45%	-11.98%	-14.47%	-22.64%	-26.58%	-30.36%
LPL	-2124.69	-2078.61	-1957.46	-1920.24	-1790.10	-1721.82	-1653.69
LBF	-	46.08	167.23	204.45	334.59	402.86	471.00

The table reports the posterior mean, standard deviation in [] and 0.95 density interval in ().

Table 4: Forecasting S&P500 overnight logarithmic realized volatility with HAR models

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4 A New Model Estimating the Daytime and Overnight Return and RV Jointly

4.1 Model Specification

The correlation coefficients between daytime and overnight returns, from Table 1, are close to zero for both SPY and QQQ, which suggests little linear correlation. Accordingly, the benchmark joint model for daytime and overnight return series treats them independently and fits two SV models, as follows:

$$r_{N,t} = \mu_N + \exp(h_{N,t}/2)u_{N,t}, \qquad u_{N,t} \sim N(0,1), \tag{15}$$

$$h_{N,t} = \alpha_N + \delta_N h_{N,t} + \sigma_{hN} v_{N,t}, \qquad v_{N,t} \sim N(0,1), \tag{16}$$

$$r_{D,t} = \mu_D + \exp(h_{D,t}/2)u_{D,t}, \qquad u_{D,t} \sim N(0,1), \tag{17}$$

$$h_{D,t} = \alpha_D + \delta_D h_{D,t-1} + \sigma_{hD}, v_{D,t} \qquad v_{D,t} \sim N(0,1). \tag{18}$$

Inasmuch as we have daytime and overnight realized volatility, we can extend the stochastic volatility model by jointly estimating the return and realized volatility as following SV-RV models (see Takahashi et al. (2009)):

$$r_{N,t} = \mu_N + \exp(h_{N,t}/2)u_{N,t}, \qquad u_{N,t} \sim N(0,1), \tag{19}$$

$$\log RV_{N,t} = \xi_N + h_{N,t} + \sigma_{RVN} e_{N,t}, \qquad e_{N,t} \sim N(0,1), \tag{20}$$

$$h_{N,t} = \alpha_N + \delta_N h_{N,t} + \sigma_{hN} v_{N,t}, \qquad v_{N,t} \sim N(0,1), \tag{21}$$

$$r_{D,t} = \mu_D + \exp(h_{D,t}/2)u_{D,t}, \qquad u_{D,t} \sim N(0,1), \tag{22}$$

$$\log RV_{D,t} = \xi_D + h_{D,t} + \sigma_{RVD} e_{D,t}, \qquad e_{D,t} \sim N(0,1), \tag{23}$$

$$h_{D,t} = \alpha_D + \delta_D h_{D,t-1} + \sigma_{hD} v_{D,t}, \qquad v_{D,t} \sim N(0,1). \tag{24}$$

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The constant term ξ (in Eq. (20) and (23)) scales the realized volatility to match the return volatility. A negative ξ (or $\exp(\xi) < 1$) indicates that the realized volatility underestimates the return volatility. Takahashi et al. (2009) fit the SV-RV model with daily close-to-close returns for the Tokyo stock price index (TOPIX) and its corresponding daytime realized volatility (to be specific, trading hour realized volatility since Tokyo Stock Exchange has a lunch break). The posterior mean of ξ is -1.0707, which implies $RV_t = 0.34 \times \exp(h_t)$. Takahashi et al. (2009) points out that the overnight (or non-trading hour) effect is much more significant than the microstructure noise in estimating realized volatility. Takahashi et al. (2009) also uses the scaling method by Hansen and Lunde (2005) to scale the daytime realized volatility to match the daily return volatility. However, either scaling the daytime realized volatility before fitting the models or adding constant term ξ in the model assumes a perfect correlation between the daytime realized volatility and overnight volatility. As we have shown that the daytime and overnight realized volatility are not perfectly correlated, scaling the daytime realized volatility could be inferior to estimating the overnight realized volatility directly.

Our SV-RV specification (Eq. (19) to (24)) treats daytime and overnight return separately, and there is no mismatch between return and corresponding realized volatility. As we will show in the following empirical results, ξ_N and ξ_D are close to zero, which supports the accuracy of our daytime and overnight realized volatility estimation.

On the other hand, both SV-RV and SV specifications treat the daytime and overnight volatility processes independently. Although the overnight and daytime returns have little linear correlation, the results in the previous section suggest a strong correlation between the daytime and overnight volatility ($h_{D,t}$ and $h_{N,t}$). To address this correlation, we propose the DN-SV-RV (day-night SV-RV) specification in Eq. (25) to (30).

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$$r_{N,t} = \mu_N + \exp(h_{N,t}/2)u_{N,t}, \qquad u_{N,t} \sim N(0,1), \tag{25}$$

$$\log RV_{N,t} = \xi_N + h_{N,t} + \sigma_{RVN}e_{N,t}, \qquad e_{N,t} \sim N(0,1), \tag{26}$$

$$h_{N,t} = \alpha_N + \beta_N h_{D,t-1} + \delta_N h_{N,t-1} + \sigma_{hN}v_{N,t}, \qquad v_{N,t} \sim N(0,1), \tag{27}$$

$$r_{D,t} = \mu_D + \exp(h_{D,t}/2)u_{D,t}, \qquad u_{D,t} \sim N(0,1), \tag{28}$$

$$\log RV_{D,t} = \xi_D + h_{D,t} + \sigma_{RVD}e_{D,t}, \qquad e_{D,t} \sim N(0,1), \tag{29}$$

$$h_{D,t} = \alpha_D + \beta_D h_{N,t} + \delta_D h_{D,t-1} + \sigma_{hD}v_{D,t}, \qquad v_{D,t} \sim N(0,1). \tag{30}$$

The major difference between this specification and the traditional SV or SV-RV approach is that the daytime latent volatility $h_{D,t}$ not only depends on its own lagged term $h_{D,t-1}$, but also on the overnight latent volatility $h_{N,t}$, and this correlation is captured by the parameter β_D in Eq. (30). Similarly, β_N in Eq. (27) captures the correlation between the lagged daytime latent volatility $h_{D,t-1}$ and the following overnight latent volatility $h_{N,t}$. Moreover, the simple SV or SV-RV model will be a special case of the DV-SV-RV model since the DN-SV-RV model will degenerate into an SV-RV model if $\beta_D = \beta_N = 0$. If there is no correlation between the daytime and overnight volatility, jointly modelling the daytime and overnight returns (and realized volatility) would be equivalent to modelling them independently.

To fit the DN-SV-RV model, it is not necessary that the daytime return/realized volatility series has the same length as that of the overnight return/realized volatility series. If both daytime and overnight series have t observations, we fit the model at the market close of day t, and we use all the past daytime and overnight information to predict the following overnight return and volatility. If the daytime series has length t while the overnight series has length $t+1$, we fit the model at the market open of day $t+1$ to predict the following daytime return and volatility. This is important for decomposing the likelihood of daytime and overnight observations and for out-of-sample prediction.

The likelihood of return and realized volatility $\{r_{D,t}, r_{N,t}, \log RV_{D,t}, \log RV_{N,t}\}_{t=1}^{T}$ of model

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M, conditional on parameter set Θ, is:

$$\begin{aligned} & p_M\left(\{r_{D,t}, r_{N,t}, \log RV_{D,t}, \log RV_{N,t}\}_{t=1}^{T} | \Theta\right) \\ =& \prod_{t=1}^{T} p_M\left(r_{D,t}, r_{N,t}, \log RV_{D,t}, \log RV_{N,t} \mid r_{D,:t-1}, r_{N,:t-1}, \log RV_{D,:t-1}, \log RV_{N,:t-1}, \Theta\right). \end{aligned} \quad (31)$$

Here $y_{:t}$ denotes $\{y_1, y_2, ...y_t\}$. For SV-RV, since the daytime and overnight volatility are modeled independently, the likelihood will be:

$$\begin{aligned} & p_M\left(\{r_{D,t}, r_{N,t}, \log RV_{D,t}, \log RV_{N,t}\}_{t=1}^{T} | \Theta\right) \\ =& \prod_{t=1}^{T} p_M\left(r_{N,t}, \log RV_{N,t} | r_{N,:t-1}, \log RV_{N,:t-1}, \Theta\right) \\ & \prod_{t=1}^{T} p_M\left(r_{D,t}, \log RV_{D,t} | r_{D,:t-1}, \log RV_{D,:t-1}, \Theta\right). \end{aligned} \quad (32)$$

However, for the DN-SV-RV model, the likelihood will be:

$$\begin{aligned} & p_M\left(\{r_{D,t}, r_{N,t}, \log RV_{D,t}, \log RV_{N,t}\}_{t=1}^{T} | \Theta\right) \\ =& \prod_{t=1}^{T} p_M\left(r_{N,t}, \log RV_{N,t} | r_{N,:t-1}, r_{D,:t-1}, \log RV_{N,:t-1}, \log RV_{D,:t-1}, \Theta\right) \\ & \prod_{t=1}^{T} p_M\left(r_{D,t}, \log RV_{D,t} | r_{D,:t-1}, r_{N,:t}, \log RV_{D,:t-1}, \log RV_{N,:t}, \Theta\right). \end{aligned} \quad (33)$$

We can see the difference between DN-SV-RV and other models in information updating. For example, at every market open, new overnight return and realized volatility observations will be available. The SV and SV-RV only update the overnight part of the model (Eq. (15) to (16) or (19) to (21)) and leave the daytime part unchanged. This represents modeling daytime series alone without overnight information, and the prediction of the daytime return and volatility is not affected even though the new overnight return and realized volatility observations are available. The DN-SV-RV model updates the joint model at every market open so that the prediction of the following daytime return and volatility will be affected by

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the newly available overnight information. To update information at the market close, when the new daytime return and realized volatility observations are available, the DN-SV-RV model update the joint model similarly.

4.2 Bayesian Estimation and Forecasting Comparison

We use Bayesian MCMC to estimate the DN-SV-RV model. We sample the daytime and overnight latent volatility process ($\{h_{D,t}\}_{t=1}^{T}$ and $\{h_{N,t}\}_{t=1}^{T}$) separately. Conditional on $\{h_{D,t}\}_{t=1}^{T}$, let Θ denote the set of parameters, the posterior distribution of $h_{N,t}$ is:

$$
\begin{aligned}
&p(h_{N,t}|r_{N,t}, \log RV_{N,t}, h_{N,-t}, h_D, \Theta) \\
\propto &p(r_{N,t}, \log RV_{N,t}|h_{N,t}, \Theta)\ p(h_{N,t+1}|h_{N,t}, h_{D,t}, \Theta)\ p(h_{D,t}|h_{N,t}, \Theta) p(h_{N,t}|h_{N,t-1}, h_{D,t-1}, \Theta) \\
\propto &\exp\left(-\frac{h_{N,t}}{2}\right) \exp\left(-\frac{(r_{N,t}-\mu_N)^2}{2\exp(h_{N,t})}\right) \exp\left(-\frac{(\log RV_{N,t}-\xi_N-h_{N,t})^2}{2\sigma_{RVN}^2}\right) \\
&\exp\left(-\frac{(h_{N,t+1}-\alpha_N-\beta_N h_{D,t}-\delta_N h_{N,t})^2}{2\sigma_{hN}^2}\right) \exp\left(-\frac{(h_{D,t}-\alpha_D-\beta_D h_{N,t}-\delta_D h_{D,t-1})^2}{2\sigma_{hD}^2}\right) \\
&\exp\left(-\frac{(h_{N,t}-\alpha_N-\beta_N h_{D,t-1}-\delta_N h_{N,t-1})^2}{2\sigma_{hN}^2}\right)
\end{aligned} \tag{34}
$$

We use the Metropolis-Hastings algorithm with tailored proposals to sample $h_{N,t}$. We need to simulate $h_{D,0}$, $h_{N,0}$ and $h_{N,T+1}$ according to Eq. (27) and (30).

While conditional on $\{h_{N,t}\}_{t=1}^{T}$, the posterior distribution of $h_{D,t}$ is:

$$
\begin{aligned}
&p(h_{D,t}|r_{D,t}, h_{D,-t}, h_N, \Theta) \\
\propto &p(r_{D,t}, \log RV_{N,t}|h_{D,t}, \Theta)\ p(h_{D,t+1}|h_{D,t}, h_{N,t+1}, \Theta)\ p(h_{N,t+1}|h_{D,t}, \Theta) p(h_{D,t}|h_{D,t-1}, h_{N,t}, \Theta) \\
\propto &\exp\left(-\frac{h_{D,t}}{2}\right) \exp\left(-\frac{(r_{D,t}-\mu_D)^2}{2\exp(h_{N,t})}\right) \exp\left(-\frac{(\log RV_{D,t}-\xi_D-h_{D,t})^2}{2\sigma_{RVD}^2}\right) \\
&\exp\left(-\frac{(h_{D,t+1}-\alpha_D-\beta_D h_{N,t+1}-\delta_D h_{D,t})^2}{2\sigma_{hD}^2}\right) \exp\left(-\frac{(h_{N,t+1}-\alpha_N-\beta_N h_{D,t}-\delta_N h_{N,t})^2}{2\sigma_{hN}^2}\right) \\
&\exp\left(-\frac{(h_{D,t}-\alpha_D-\beta_D h_{N,t}-\delta_D h_{D,t-1})^2}{2\sigma_{hD}^2}\right)
\end{aligned} \tag{35}
$$

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Similarly, we use the Metropolis-Hastings algorithm with tailored proposals to sample $h_{D,t}$. $h_{D,0}$, $h_{N,T+1}$ and $h_{D,T+1}$ are simulated accordingly. If we are fitting the DN-SV-RV model at the market open time, $h_{N,T+1}$ will be available and we only need to simulate $h_{D,0}$ and $h_{D,T+1}$.

Notice that Eq. (27) and (30) suggest a bi-variate autoregression process of $[h_{N,t}, h_{D,t}]'$. Denote the transition matrix $\left(\begin{smallmatrix} \delta_N & \beta_N \\ \beta_D & \delta_D \end{smallmatrix}\right)$ as Φ. Stationary condition requires that the eigenvalues of matrix Φ are smaller than 1 in modulus. For the SV-RV/SV model, we need $|\delta_N| < 1$ and $|\delta_D| < 1$ for stationary, which is a special case sine δ_N and δ_N are eigenvalues of Φ when $\beta_N = \beta_D = 0$. Given the noise term in Eq. (27) and (30) are independent, conditional on $\{h_{N,t}\}_{t=1}^T$ and $\{h_{D,t}\}_{t=1}^T$, we draw α_N, β_N and δ_N independently from α_D, β_D and δ_D by assuming independent normal priors. We use the rejection method to constrain the parameter draws of Φ within the stationary region.

We show the MCMC iteration steps and parameter prior distributions for the DN-SV-RV model here. The SV-RV/SV model can be treated as a nested model of the DN-SV-RV model, and the parameters have the same prior distributions as the DN-SV-RV model:

- $\mu_N, \mu_D \mid h_N, r_N, h_D, r_D$: Conjugate normal prior distributions $N(0, 5)$.
- $h_D \mid h_N, r_D, \log RV_D$: Metropolis-Hasting with tailored proposal distributions.
- $h_N \mid h_D, r_N, \log RV_N$: Metropolis-Hasting with tailored proposal distributions.
- $\alpha_N, \beta_N, \delta_N, \alpha_D, \beta_D, \delta_D \mid h_N, h_D$: Conjugate normal prior distributions $N(0, 5)$ with stationary constraint on Φ.
- $\xi_N, \xi_D \mid h_N, h_D, \log RV_D, \log RV_N$ Conjugate normal prior distributions $N(0, 5)$.
- $\sigma_{hN}, \sigma_{hd}, \sigma_{RVN}, \sigma_{RVD} \mid h_N, h_D, \log RV_D, \log RV_N$: Conjugate inverse gamma prior distributions $IG(\frac{3}{2}, \frac{0.5}{2})$.

We set flat prior distributions for all parameters to avoid subjective bias. Also, we set the total number of MCMC iterations to be 100,000, and we keep 1 of every 10 draws. We also

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set 5,000 burn-in iterations for parameter convergence.

To compare the out-of-sample performance, we focus on forecasting the daytime and overnight return distributions. We use the following logarithmic predictive likelihood of model M:

$$LPL_M = \sum_{t=s}^{T-1} (\log p_M(r_{N,t+1}, r_{D,t+1} | r_{D,:t}, r_{N,:t}, \log RV_{D,:t}, \log RV_{N,:t})) \tag{36}$$

Given our sample size $T = 2686$, we use the initial $s = 500$ observation to initially fit the models. Moreover, we can further decompose the predictive likelihood in Eq. (36) into daytime and overnight parts, as follows:

$$\begin{aligned} & p_M(r_{N,t+1}, r_{D,t+1} | r_{D,:t}, r_{N,:t}, \log RV_{D,:t}, \log RV_{N,:t}) \\ = & \, p_M(r_{N,t+1} | r_{D,:t}, r_{N,:t}, \log RV_{D,:t}, \log RV_{N,:t}) \\ & \, p_M(r_{D,t+1} | r_{D,:t}, r_{N,:t+1}, \log RV_{D,:t}, \log RV_{N,:t+1}) \end{aligned} \tag{37}$$

Then the logarithmic predictive likelihood in Eq. (36) can be decomposed into overnight and daytime parts, as follows:

$$LPL_{N,M} = \sum_{t=s}^{T-1} (\log p_M(r_{N,t+1} | r_{D,:t}, r_{N,:t}, \log RV_{D,:t}, \log RV_{N,:t})) \tag{38}$$

$$LPL_{D,M} = \sum_{t=s}^{T-1} (\log p_M(r_{D,t+1} | r_{D,:t}, r_{N,:t+1}, \log RV_{D,:t}, \log RV_{N,:t+1})) \tag{39}$$

We have $LPL_M = LPL_{N,M} + LPL_{D,M}$ given conditional independence.

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For the SV-RV model, the predictive likelihood is estimated as:

$$\begin{aligned}
&p_M(r_{N,t+1}|r_{D,:t}, r_{N,:t}, \log RV_{D,:t}, \log RV_{N,:t}) \\
&= p_M(r_{N,t+1}|r_{N,:t}, \log RV_{N,:t}) \\
&\approx \frac{1}{N}\sum_{i=1}^{N} f_N\left(r_{N,t+1}|\mu_N^{(i)},\ \exp(\alpha_N^{(i)} + \delta_N^{(i)} h_{N,t}^{(i)} + \sigma_{hN}^{(i)} v_{N,t}^{(i)})\right)
\end{aligned} \tag{40}$$

and

$$\begin{aligned}
&p_M(r_{D,t+1}|r_{D,:t}, r_{N,:t+1}, \log RV_{D,:t}, \log RV_{N,:t+1}) \\
&= p_M(r_{D,t+1}|r_{D,:t}, \log RV_{D,:t}) \\
&\approx \frac{1}{N}\sum_{i=1}^{N} f_N\left(r_{D,t+1}|\mu_D^{(i)},\ \exp(\alpha_D^{(i)} + \delta_D^{(i)} h_{D,t}^{(i)} + \sigma_{hD}^{(i)} v_{D,t}^{(i)})\right)
\end{aligned} \tag{41}$$

Here $\mu_N^{(i)}$, $\alpha_N^{(i)}$, $\delta_N^{(i)}$, $h_{N,t}^{(i)}$ and $\sigma_{hN}^{(i)}$ are MCMC draws by fitting the overnight part of the SV-RV model with overnight returns $\{r_{N,i}\}_{i=1}^{t}$ and overnight realized volatility $\{\log RV_{N,i}\}_{i=1}^{t}$ jointly. Similarly, $\mu_N^{(i)}$, $\alpha_D^{(i)}$, $\delta_D^{(i)}$, $h_{D,t}^{(i)}$ and $\sigma_{hD}^{(i)}$ are MCMC draws by fitting the daytime part of the SV-RV model with daytime return $\{r_{D,i}\}_{i=1}^{t}$ and daytime realized volatility $\{\log RV_{D,i}\}_{i=1}^{t}$. $v_{N,t}^{(i)}$ and $v_{D,t}^{(i)}$ are iid draws from a standard normal distribution. For the SV model, it is the same as Eq. 40 and 41 to evaluate the logarithmic predictive likelihood. However, we fit the SV model without realized volatility.

For the DN-SV-RV model, the predictive likelihood is estimated as:

$$\begin{aligned}
&p_M(r_{N,t+1}|r_{D,:t}, r_{N,:t}, \log RV_{D,:t}, \log RV_{N,:t}) \\
&\approx \frac{1}{N}\sum_{i=1}^{N} f_N\left(r_{N,t+1}|\mu_N^{(i)},\ \exp(\alpha_N^{(i)} + \beta_N^{(i)} h_{D,t}^{(i)} + \delta_N^{(i)} h_{N,t}^{(i)} + \sigma_{hN}^{(i)} v_N^{(i)})\right)
\end{aligned} \tag{42}$$

$$\begin{aligned}
&p_M(r_{D,t+1}|r_{D,:t}, r_{N,:t+1}, \log RV_{D,:t}, \log RV_{N,:t+1}) \\
&\approx \frac{1}{N}\sum_{i=1}^{N} f_N\left(r_{D,t+1}|\mu_D^{(i)},\ \exp(\alpha_D^{(i)} + \beta_D^{(i)} h_{N,t+1}^{(i)} + \delta_D^{(i)} h_{D,t}^{(i)} + \sigma_{hD}^{(i)} v_D^{(i)})\right)
\end{aligned} \tag{43}$$

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$\mu_N^{(i)}$, $\alpha_N^{(i)}$, $\beta_N^{(i)}$, $\delta_N^{(i)}$, $h_{D,t}^{(i)}$, $h_{N,t}^{(i)}$ and $\sigma_{hN}^{(i)}$ are MCMC draws from fitting the DN-SV-RV with $\{r_{N,i}\}_{i=1}^{t}$, $\{r_{D,i}\}_{i=1}^{t}$, $\{\log RV_{N,i}\}_{i=1}^{t}$ and $\{\log RV_{D,i}\}_{i=1}^{t}$. This step is to predict $r_{N,t+1}$ given the information available at market close on day t. On the next day, when the market is open, $r_{N,t+1}$ and $\log RV_{N,t+1}$ are available. Then we fit the DN-SV-RV model with $\{r_{N,i}\}_{i=1}^{t+1}$, $\{r_{D,i}\}_{i=1}^{t}$, $\{\log RV_{N,i}\}_{i=1}^{t+1}$ and $\{\log RV_{D,i}\}_{i=1}^{t}$ and draw $\mu_D^{(i)}$, $\alpha_D^{(i)}$, $\beta_D^{(i)}$, $\delta_D^{(i)}$, $h_{D,t}^{(i)}$, $h_{N,t+1}^{(i)}$ and $\sigma_{hD}^{(i)}$ accordingly.

Table 5 summarizes the estimation results for SV (Eq (15) to (18)), SV-RV (Eq (19) to (24)) and DN-SV-RV (Eq (25) to (30)) for SPY and QQQ. As we have stated before, the key difference between DN-SV-RV and the traditional SV-RV or SV is β_D and β_N. The estimation results support our expectation that the correlation between daytime and overnight volatility is strong. Taking the SPY result as an example, $\beta_D = 0.6171$, which indicates a strong correlation between overnight volatility and following daytime volatility. Similarly, $\beta_N = 0.3665$ indicates a significant correlation between daytime volatility and following overnight volatility. Moreover, if the daytime or overnight return series is modelling with SV or SV-RV independently, we find robust volatility clustering as δ_D and δ_N are around 0.95. However, as we have noted, δ_D only captures the correlation between today's daytime volatility and yesterday's daytime volatility, and there is an overnight gap between them. Incorporating the overnight volatility into the daytime volatility equation (incorporating the daytime volatility into the overnight volatility equation) will significantly decrease the auto-correlation coefficient $\delta_D = 0.3016$ ($\delta_N = 0.5877$).

After controlling the overnight (daytime) volatility, the auto-correlation of the daytime (overnight) volatility is much weaker. However, there is no contradiction to the well-documented volatility clustering. The daytime and overnight volatility, like the daytime and overnight returns, are the converse of each other. Although the overnight volatility has a different long-run mean and variance from that of the daytime volatility (overnight volatility is smaller), they are driven by the same risk factor, and the volatility clustering should be persistent throughout the day and night. When predicting the daytime volatility,

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the most recent information is not the lagged daytime volatility, but the overnight volatility and vice versa. Previous studies that focus on daytime volatility alone use only a subset of the volatility information. The overnight realized volatility and the DV-SV-RV complete those overnight gaps.

Moreover, from Table 5, the volatility-adjusted mean of overnight returns is lower than that of daytime returns. Back to Table 1, the simple mean of overnight returns (0.0268% for SPY and 0.0478% for QQQ) is higher than that of the daytime returns (0.0218% for SPY and 0.0233% for QQQ). However, the mean parameters in the DN-SV-RV model indicate that the overnight return mean ($\mu_N = 0.0525\%$ for SPY and $\mu_N = 0.0772\%$ for QQQ) is lower the than daytime return mean ($\mu_D = 0.0904\%$ for SPY and $\mu_D = 0.1275\%$ for QQQ). The stochastic volatility could be a potential explanation of the observed high overnight returns.

The last four rows of Table 5 further compares the out-of-sample forecasting of the daytime and overnight return distributions, and we find strong supportive evidence for the overnight realized volatility and our DN-SV-RV model. Take SPY as an example, comparing SV-RV and SV. We see the realized volatility, especially the overnight realized volatility estimated from the high-frequency futures prices, can significantly improve the forecasts of the return distribution. The daytime (overnight) logarithmic predictive Bayes factor of the SV-RV model relative to the SV model is 40.82 (50.93). The logarithmic predictive Bayes factor strongly supports the value of our realized volatility estimated from the high-frequency futures prices. The realized volatility, as a consistent volatility measure, improves the future return distribution forecasting. The improvement shows that our overnight realized volatility not only enhances the daytime realized volatility forecasting in the HAR-RV model, which models the realized volatility process itself, but also matches the corresponding overnight returns. The improvement of the SV-RV model compared to the simple SV model confirms that the overnight realized volatility provides volatility information that helps the SV-RV model identify the real overnight volatility process and achieve better return prediction.

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Moreover, the DN-SV-RV model further exploits the correlation between daytime and overnight volatility process. The daytime (overnight) logarithmic Bayes factor of the DN-SV-RV model over the SV-RV model is 20 (44.11) for SPY. From SV-RV to DN-SV-RV, we do not introduce extra data and the SV-RV is a nested version of DN-SV-RV. As discussed earlier, the DN-SV-RV model leads to significantly weaker auto-correlation in the daytime and overnight volatility processes, but strong inter-correlation. Significant forecasting improvements can be achieved by modeling the correlation between daytime and overnight volatility. Again, the consistent results of QQQ, as shown in Table 5, support the robustness of our findings.

The out-of-sample results confirm that it is necessary to incorporate the overnight (daytime) information for predicting the following daytime (overnight) volatility. The results here are consistent with Section 3's findings that overnight (daytime) realized volatility can significantly improve the following daytime (overnight) realized volatility.

We have shown that the overnight volatility process is not a negligible noise process for major U.S. equity indexes (S&P and NASDAQ). The volatility clustering is consistent throughout day and night and overnight volatility contains non-negligible information for volatility modeling and forecasting. To corroborate our findings, in the following section, we further examine whether the daily close-to-close realized volatility RV_t (estimated from the almost 24-hour traded E-mini index futures) dominates the summation of daytime realized volatility and squared overnight returns as a measure of daily volatility.

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	SPY			QQQ		
	SV	SV-RV	DN-SV-RV	SV	SV-RV	DN-SV-RV
μ_N	0.0445	0.0501	0.0525	0.0670	0.0731	0.0772
	[0.007]	[0.007]	[0.007]	[0.009]	[0.009]	[0.008]
	(0.030, 0.059)	(0.036, 0.064)	(0.039, 0.066)	(0.050, 0.084)	(0.057, 0.090)	(0.061, 0.093)
α_N	-0.0640	-0.0875	-0.2021	-0.0854	-0.0928	-0.2714
	[0.015]	[0.014]	[0.024]	[0.019]	[0.014]	[0.028]
	(-0.094, -0.037)	(-0.115, -0.061)	(-0.252, -0.156)	(-0.128, -0.052)	(-0.120, -0.067)	(-0.328, -0.219)
δ_N	0.9597	0.9460	0.5877	0.9350	0.9311	0.5718
	[0.008]	[0.007]	[0.029]	[0.013]	[0.009]	[0.031]
	(0.942, 0.975)	(0.931, 0.960)	(0.530, 0.642)	(0.907, 0.958)	(0.914, 0.947)	(0.510, 0.628)
β_N	-	-	0.3665	-	-	0.3922
			[0.027]			[0.031]
			(0.315, 0.422)			(0.335, 0.455)
σ_N	0.3104	0.3252	0.3691	0.3585	0.3474	0.4009
	[0.027]	[0.013]	[0.014]	[0.035]	[0.014]	[0.015]
	(0.261, 0.366)	(0.300, 0.352)	(0.342, 0.398)	(0.295, 0.431)	(0.320, 0.376)	(0.371, 0.431)
μ_D	0.0616	0.0780	0.0904	0.0796	0.1098	0.1275
	[0.009]	[0.008]	[0.008]	[0.012]	[0.011]	[0.011]
	(0.044, 0.079)	(0.062, 0.095)	(0.074, 0.106)	(0.057, 0.103)	(0.087, 0.132)	(0.106, 0.149)
α_D	-0.0689	-0.1176	0.1087	-0.0452	-0.0899	0.1849
	[0.015]	[0.016]	[0.034]	[0.011]	[0.013]	[0.033]
	(-0.101, -0.041)	(-0.149, -0.087)	(0.043, 0.176)	(-0.069, -0.024)	(-0.116, -0.065)	(0.121, 0.249)
δ_D	0.9436	0.9085	0.3016	0.9382	0.8864	0.3567
	[0.010]	[0.010]	[0.031]	[0.012]	[0.012]	[0.032]
	(0.921, 0.962)	(0.888, 0.928)	(0.240, 0.363)	(0.914, 0.959)	(0.863, 0.909)	(0.295, 0.421)
β_D	-	-	0.6171	-	-	0.5116
			[0.033]			[0.031]
			(0.554, 0.681)			(0.449, 0.572)
σ_D	0.3628	0.4207	0.4871	0.3458	0.4195	0.4842
	[0.031]	[0.016]	[0.010]	[0.032]	[0.017]	[0.011]
	(0.306, 0.425)	(0.390, 0.453)	(0.467, 0.506)	(0.285, 0.410)	(0.388, 0.453)	(0.461, 0.506)
ξ_N	-	0.1264	0.1409	-	-0.0108	0.0084
		[0.028]	[0.029]		[0.029]	[0.030]
		(0.070, 0.179)	(0.083, 0.198)		(-0.070, 0.045)	(-0.051, 0.066)
σ_{RVN}	-	0.3536	0.2948	-	0.3619	0.2915
		[0.010]	[0.014]		[0.010]	[0.015]
		(0.335, 0.372)	(0.267, 0.320)		(0.342, 0.382)	(0.259, 0.320)
ξ_D	-	0.1278	0.1371	-	0.0452	0.0522
		[0.030]	[0.028]		[0.028]	[0.028]
		(0.069, 0.187)	(0.082, 0.191)		(-0.011, 0.100)	(-0.002, 0.109)
σ_{RVD}	-	0.3219	0.1650	-	0.3175	0.1811
		[0.013]	[0.018]		[0.014]	[0.021]
		(0.295, 0.347)	(0.133, 0.202)		(0.288, 0.344)	(0.144, 0.223)
LPL_D	-1869.05	-1828.23	**-1808.23**	-2422.41	-2390.76	**-2358.84**
LPL_N	-1425.03	-1374.10	**-1329.99**	-1790.51	-1753.50	**-1704.19**
$LPL_D + LPL_N$	-3294.07	-3202.33	**-3138.22**	-4212.92	-4144.26	**-4063.03**

The table reports the posterior mean, standard deviation in [] and 0.95 density interval in ().

Table 5: The estimation results for the SV, SV-RV and DN-SV-RV models.

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5 The Daily Close-to-Close Realized Volatility

5.1 Joint Models for Daily Close-to-Close Return and Volatility Measures

To measure the daily close-to-close volatility, we have the following three candidates of daily realized volatility measure:

- $RV_{D,t}$: The daytime RV, estimated from the high-frequency prices within main trading hours, which overlooked the overnight trading information.
- RV_t: The daily close-to-close RV, estimated from the almost 24-hour high-frequency future prices. Given our estimation of realized measures, the daily close-to-close realized volatility is the summation of daytime and overnight realized volatility. ($RV_t = RV_{N,t} + RV_{D,t}$).
- NRV_t: The daytime RV plus the squared overnight returns ($NRV_t = RV_{D,t} + r_{N,t}^2$).

We jointly estimate the daily close-to-close return r_t with different volatility measure VM_t (could be $RV_{D,t}$ RV_t or NRV_t), as the following SV-RV model (Takahashi et al. 2009):

$$r_t = \mu + exp(h_t)u_t, \qquad u_t \sim N(0,1), \tag{44}$$

$$\log VM_t = \xi + h_t + \sigma_{VM} e_t \qquad e_t \sim N(0,1), \tag{45}$$

$$h_t = \alpha + \delta h_{t-1} + \sigma_h v_t, \qquad v_t \sim N(0,1). \tag{46}$$

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Moreover, we also jointly estimate daily returns and all volatility measures as follows:

$$
\begin{aligned}
r_t &= \mu + exp(h_t)u_t, & u_t &\sim N(0,1),\\
\log RV_{D,t} &= \xi_{RVD} + h_t + \sigma_{RVD}e_{RVD,t}, & e_{RVD,t} &\sim N(0,1)\\
\log RV_t &= \xi_{RV} + h_t + \sigma_{RV}e_{RV,t}, & e_{RV,t} &\sim N(0,1)\\
\log NRV_t &= \xi_{NRV} + h_t + \sigma_{NRV}e_{NRV,t}, & e_{NRV,t} &\sim N(0,1),\\
h_t &= \alpha + \delta h_{t-1} + \sigma_h v_t, & v_t &\sim N(0,1).
\end{aligned}
$$

where all three volatility measures are jointly modeled with the latent daily close-to-close volatility h_t.

When fitting the SV-RV model with r_t and $\log VM_t$, the parameter ξ corrects the bias between the mean of logarithmic volatility measure ($\log VM_t$) and the mean of the estimated latent logarithmic daily return volatility process (h_t). As we have shown in previous sections, we expect ξ_{RVD} to be negative, since the $RV_{D,t}$ omitted the overnight volatility. For RV_t (NRV_t), we expect ξ_{RV} (ξ_{NRV}) to be close to zero.

Another method of measuring daily close-to-close to realized volatility is to scale the daytime realized volatility with parameter $c = \frac{(r_t - \bar{r}_t)^2}{\sum_t RV_{D,t}}$ to match the variance of daily close-to-close returns. However, the scaled $RV_{D,t}$ omits the overnight information, and the scaling parameter c is captured by the parameter ξ in the SV-RV model[4].

On the other hand, the parameter σ_{VM} in Eq. (45) measures the noise of VM_t, which is captured by the estimated latent volatility process h_t. When jointly estimating daily closed-to-closed return r_t and the daytime realized volatility $RV_{D,t}$, we expect not only significantly lower ξ_{RVD}, but also significantly higher σ_{RVD}. The noise should not be solely attributed to measurement errors during the estimation of $RV_{D,t}$. Unless the overnight volatility process is perfectly correlated with the daytime volatility process, which is not the case given the empirical results of our proposed DN-SV-RV model, there should be significant mismatch

[4]Jointly estimating r_t and $cRV_{D,t}$ leads to almost identical empirical result as that of estimating r_t and $RV_{D,t}$ with SV-RV model.

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between the dynamics of the daily close-to-close volatility and open to close processes, which should lead to significantly higher σ_{RVD}.

Recall $NRV_t = RV_{D,t} + r_{N,t}^2$ and $RV_t = RV_{D,t} + RV_{N,t}$. If the overnight volatility measures, either $RV_{N,t}$ or $r_{N,t}^2$, are purely noise or too noisy as volatility measures, we expect significantly higher σ_{VM}. On the contrary, if the our overnight realized volatility estimated from the E-mini futures accurately measures the overnight volatility process, we expect that σ_{RV} will be significantly smaller. In other words, greater portion of the variation of the logarithmic volatility measure process is expected to explained by the h_t process, which is measured by $1 - \sigma_{VM}^2/Var(\log VM_t)$.

In addition, we compare the following logarithmic predictive likelihood as in Section 4.2:

$$\begin{aligned} LPL_M &= \sum_{t=s}^{T-1} \log p_M(r_{t+1}|r_{:t}, \log VM_{:t}) \\ &\approx \sum_{t=s}^{T-1} \log \left(\frac{1}{N} \sum_{i=1}^{N} f_N \left(r_{t+1}|\mu^{(i)}, \ \exp(\alpha^{(i)} + \delta^{(i)} h_t^{(i)} + \sigma_h^{(i)} v^{(i)}) \right) \right) \end{aligned} \tag{47}$$

The logarithmic predictive likelihood measures the density forecasting of future daily close-to-close returns. If jointly estimating r_t and $\log VM_t$ lead to a latent volatility process h_t that only captures the dynamics of $\log VM_t$, we would expect that the logarithmic predictive likelihood of r_t will be significantly lower. Otherwise, VM_t provides incremental information for identifying the real daily close-to-close return volatility and improve the density forecasting of daily returns. We estimate SV-RV models with similar MCMC steps as in Section 4.2. Given the MCMC draws, the logarithmic predictive likelihood of future returns can be estimated as in Eq. (47).

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5.2 The Empirical Results for Daily Close-to-Close Return and Volatility Measures

We analyze the empirical results for SPY as example in this section. The results for QQQ are consistent with what we find for SPY, and strengthen the robustness of conclusions. We first report the summary statistics of daily return and volatility measures in Table 6. The mean of RV_t is 1.1576, which is close to the variance of daily return r_t. On the contrary, the mean of $RV_{D,t}$ is 0.6704, the gap between the mean of $RV_{D,t}$ and the variance of r_t is fulfilled by $RV_{N,t}$.

	r_t	RV_t	NRV_t	$RV_{D,t}$	$\log RV_t$	$\log NRV_t$	$\log RV_{D,t}$
SPY							
Mean	0.0486	1.1576	1.1813	0.6704	-0.5768	-0.7378	-1.1555
Var	1.1361	10.9895	25.9130	4.0952	1.0335	1.2763	1.1008
Skewness	-0.7053	11.4391	23.4007	12.9468	0.7371	0.6428	0.6499
Kurtosis	14.0259	162.2479	737.6325	220.0537	1.3477	1.0832	1.2005
QQQ							
Mean	0.0711	1.3933	1.4492	0.8718	-0.2675	-0.3607	-0.7453
Var	1.4369	11.2916	26.3825	4.5856	0.8745	1.0420	0.9100
Skewness	-0.5674	10.6151	20.1174	11.9575	0.7453	0.6753	0.6513
Kurtosis	9.1864	142.2328	522.8667	194.9334	1.4663	1.2757	1.2889

Sample period: 2009-10-29 to 2020-07-01, totally 2686 observations.

Table 6: Summary statistics of daily return and volatility measures.

The estimation results are summarized in Table 7. We confirm that $RV_{D,t}$ significantly underestimate the daily close-to-close return volatility as $\xi = -0.4720$, which is consistent with the fact that over 40% daily volatility is overnight volatility for U.S. equity indexes. As expected, ξ is 0.1092 (0.0384) for RV_t (NRV_t), which suggests that RV_t and NRV_t fill the overnight volatility gap. Additionally, the residual for RV_t is significantly smaller compared to that of NRV_t and $RV_{D,t}$. Fitting the SV-RV models with SPY data, the estimated $1 - \sigma_{VM}^2/\widehat{Var}(\log VM_t)$ is 92.9%, 84.0% and 88.0% for RV_t, NRV_t and $RV_{D,t}$ respectively.

Consistent with previous discussion, squared overnight returns $r_{N,t}^2$, as a measure of overnight volatility, is too noisy so that NRV_t has the greatest noise parameter $\sigma_{VM} = 0.4516$.

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	SPY				QQQ			
	RV_t	NRV_t	$RV_{D,t}$	All	RV_t	NRV_t	$RV_{D,t}$	All
μ	0.1195	0.1089	0.1359	0.1404	0.1741	0.1568	0.1900	0.2038
	[0.011]	[0.011]	[0.012]	[0.011]	[0.015]	[0.014]	[0.015]	[0.014]
	(0.097, 0.142)	(0.088, 0.130)	(0.113, 0.159)	(0.119, 0.162)	(0.145, 0.203)	(0.130, 0.184)	(0.161, 0.220)	(0.176, 0.231)
α	-0.0441	-0.0748	-0.0488	-0.0848	-0.0213	-0.0443	-0.0228	-0.0408
	[0.009]	[0.013]	[0.010]	[0.012]	[0.008]	[0.011]	[0.008]	[0.011]
	(-0.061, -0.027)	(-0.102, -0.050)	(-0.068, -0.031)	(-0.108, -0.062)	(-0.036, -0.007)	(-0.067, -0.023)	(-0.038, -0.007)	(-0.062, -0.020)
δ	0.9359	0.9040	0.9287	0.8777	0.9201	0.8759	0.9119	0.8487
	[0.008]	[0.012]	[0.009]	[0.009]	[0.009]	[0.015]	[0.010]	[0.010]
	(0.921, 0.951)	(0.880, 0.926)	(0.912, 0.945)	(0.859, 0.896)	(0.902, 0.937)	(0.844, 0.904)	(0.892, 0.931)	(0.829, 0.869)
σ_h	0.3471	0.4496	0.3684	0.4866	0.3524	0.4518	0.3654	0.4918
	[0.012]	[0.022]	[0.014]	[0.007]	[0.012]	[0.025]	[0.015]	[0.007]
	(0.325, 0.371)	(0.407, 0.495)	(0.342, 0.397)	(0.473, 0.501)	(0.328, 0.377)	(0.405, 0.504)	(0.337, 0.394)	(0.478, 0.506)
ξ_{RVD}	-	-	-0.4720	-0.4615	-	-	-0.4888	-0.4760
			[0.029]	[0.028]			[0.029]	[0.027]
			(-0.528, -0.416)	(-0.517, -0.409)			(-0.545, -0.431)	(-0.529, -0.422)
σ_{RVD}	-	-	0.3638	0.2374	-	-	0.3618	0.2216
			[0.011]	[0.004]			[0.012]	[0.004]
			(0.341, 0.386)	(0.229, 0.246)			(0.338, 0.384)	(0.214, 0.229)
ξ_{RV}	0.1092	-	-	0.1172	-0.0019	-	-	0.0018
	[0.028]			[0.028]	[0.029]			[0.027]
	(0.054, 0.164)			(0.063, 0.169)	(-0.057, 0.055)			(-0.050, 0.055)
σ_{RV}	0.2705	-	-	0.1161	0.2695	-	-	0.1025
	[0.010]			[0.005]	[0.011]			[0.005]
	(0.250, 0.290)			(0.105, 0.127)	(0.248, 0.290)			(0.093, 0.112)
ξ_{NRV}	-	0.0384	-	-0.0438	-	-0.0068	-	-0.0913
		[0.030]		[0.029]		[0.031]		[0.028]
		(-0.020, 0.098)		(-0.101, 0.010)		(-0.068, 0.053)		(-0.145, -0.037)
σ_{NRV}	-	0.4516	-	0.4333	-	0.4406	-	0.4136
		[0.015]		[0.006]		[0.017]		[0.006]
		(0.421, 0.480)		(0.421, 0.446)		(0.405, 0.472)		(0.402, 0.425)
LPL	-2409.93	-2422.67	-2408.38	-2412.87	-2902.76	-2920.64	-2905.02	-2905.06

The table reports the posterior mean, standard deviation in [] and 0.95 density interval in ().

Table 7: Estimating daily close-to-close return and volatility measure(s).

The prior distributions are: $p(\mu) \sim N(0,5)$, $p(\alpha) \sim N(0,5)$ $p(\xi) \sim N(0,5)$, $p(\sigma^2_{RV}) \sim IG(3/2, 0.5/2)$ and $p(\sigma^2_h) \sim IG(3/2, 0.5/2)$. $p(\delta) \sim N(0.9,5)I_{|\delta|<1}$, a normal distribution truncated to the stationary region (-1,1). The LPL of a simple SV model is -2469.98 (-2957.66) for SPY (QQQ).

For $RV_{D,t}$, the estimated noise parameter $\sigma_{VM} = 0.3638$ seems to be a result of the mismatch between the daily close-to-close and daytime volatility process. By completing the $RV_{D,t}$ with overnight realized volatility $RV_{N,t}$, the daily close-to-close realized volatility RV_t has smallest noise parameter $\sigma_{VM} = 0.2705$. Since 92.9% percent of the variations of log RV_t can be captured by the estimated daily logarithmic volatility process h_t, the empirical results support that RV_t out-performs NRV_t and $RV_{D,t}$, as a measure of daily volatility.

The conclusion is consistent when jointly modeling daily return and all three volatility

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measures. The joint estimation identifies a rough latent volatility process as we observe weaker auto-correlation ($\delta = 0.8777$) and higher noise ($\sigma_h = 0.4866$) in the latent volatility process (h_t). The identified h_t process almost fully matches the variations of the $\log RV_t$ process given $1 - \sigma^2_{RV}/\widehat{Var}(\log RV_t) = 98.7\%$. For $RV_{D,t}$, the noise portion is higher than that of RV_t as $1 - \sigma^2_{RVD}/\widehat{Var}(\log RV_{D,t}) = 94.9\%$. NRV_t has the greatest portion of noise compared to RV_t and $RV_{D,t}$ given $1 - \sigma^2_{NRV}/\widehat{Var}(\log NRV_t) = 85.3\%$.

The potential concern against the superiority of RV_t is that the estimated h_t volatility process maximizes the likelihood of $\log RV_t$ instead of r_t, so that the estimated noise σ_{RV} is small. The last row of Table 7 compare the logarithm predictive likelihood of future returns (y_{t+1}). We document that jointly estimating r_t and volatility measures(s) improves the identification and prediction of the daily close-to-close return volatility process (h_t) compared to the simple SV model without volatility measures. Among all the volatility measures, NRV_t has the worst predictive performance, which further confirms the in-sample results that σ_{NRV} is significantly higher. As we have discussed, given $NRV_t = RV_{D,t} + r^2_{N,t}$, the squared overnight return $r_{N,t}$ could be too noisy as a measure of overnight volatility process. For RV_t and $RV_{D,t}$, there is no significant difference between them in return density forecasting.

The goal of this paper is not to look for the best daily volatility forecasting model specification. We use the out-of-sample forecasting comparisons to show that our proposed RV_t dominate NRV_t in exploiting the overnight volatility information. The out-of-sample results strengthen the in-sample estimation results that even the SV-RV model, fitting r_t and RV_t jointly, identifies a latent volatility process that almost identical to the logarithmic volatility measure process $\log RV_t$, the model still has one of the best out-of-sample return density forecasting.

Given the empirical results, we conclude that the realized volatility measures RV_t, estimated from the almost 24-hour traded index future prices, dominate the squared overnight returns in measuring the overnight volatility. By filling the overnight volatility gap, RV_t

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also significantly improves the daytime realized volatility in matching the real daily close-to-close volatility process. As an accurate daily volatility measure, RV_t significantly contributes to the existing literature in volatility modeling and forecasting, portfolio management, risk management, etc.

6 Conclusion

The stock market never sleeps; in fact, "it loves the night". We document that after-trading-hour return and volatility are significant for the U.S. equity market as the overnight volatility comprises about 40% of the daily close-to-close volatility. It is sub-optimal to overlook the overnight volatility.

In this paper, we select equity index future prices to capture the daytime and overnight volatility, as the futures market trades for almost the full day. Using futures high frequency data, we estimate both the daytime and overnight realized volatility. We demonstrate that overnight (daytime) realized volatility will significantly improve the forecasting of the following daytime (overnight) realized volatility. The inter-correlation between daytime and overnight realized volatility is economically significant and informative for realized volatility prediction. Inspired by the relationship between daytime and overnight volatility, we propose the stochastic volatility model by jointly estimating overnight and daytime returns and their realized volatility, as in the DN-SV-RV model. This model significantly improves both daytime and overnight return density forecasting. The out-of-sample improvements strongly support the value of the overnight realized volatility estimated from the equity futures market. Also, the estimation outcome provides evidence of the non-negligible importance of cross-correlation between daytime and overnight volatility. The evidence suggests that daytime and overnight volatility complement one another. We conclude that volatility clustering is persistent throughout the day and night. Moreover, our proposed daily close-to-close realized volatility accurately measures both daytime and overnight realized volatility. Further

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applications of the daily close-to-close RV are promising for both academic and practical perspectives.

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7 Appendix

7.1 Supplementary Overnight Return Plots

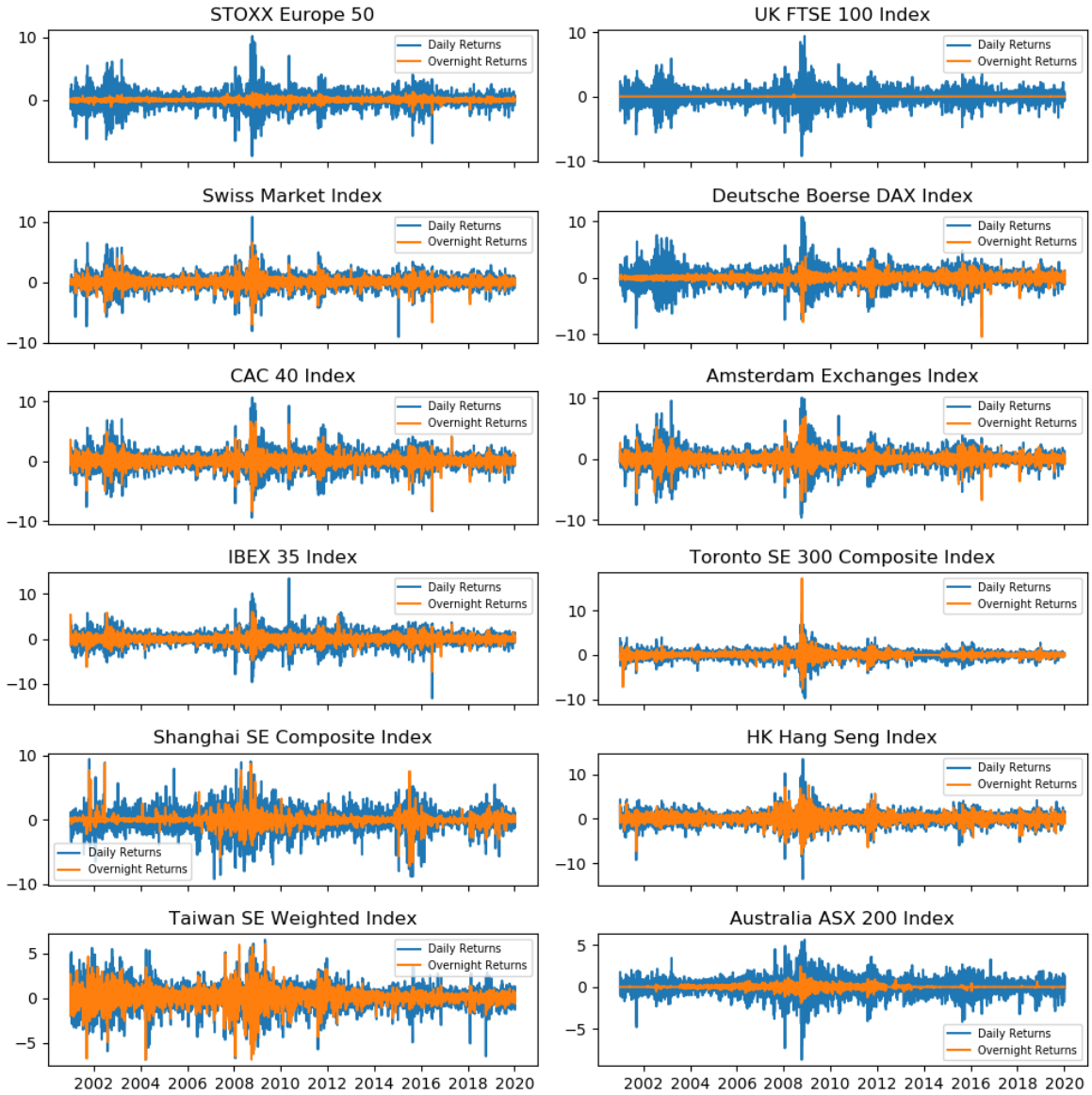


Figure 5: Overnight returns of major global market indexes.

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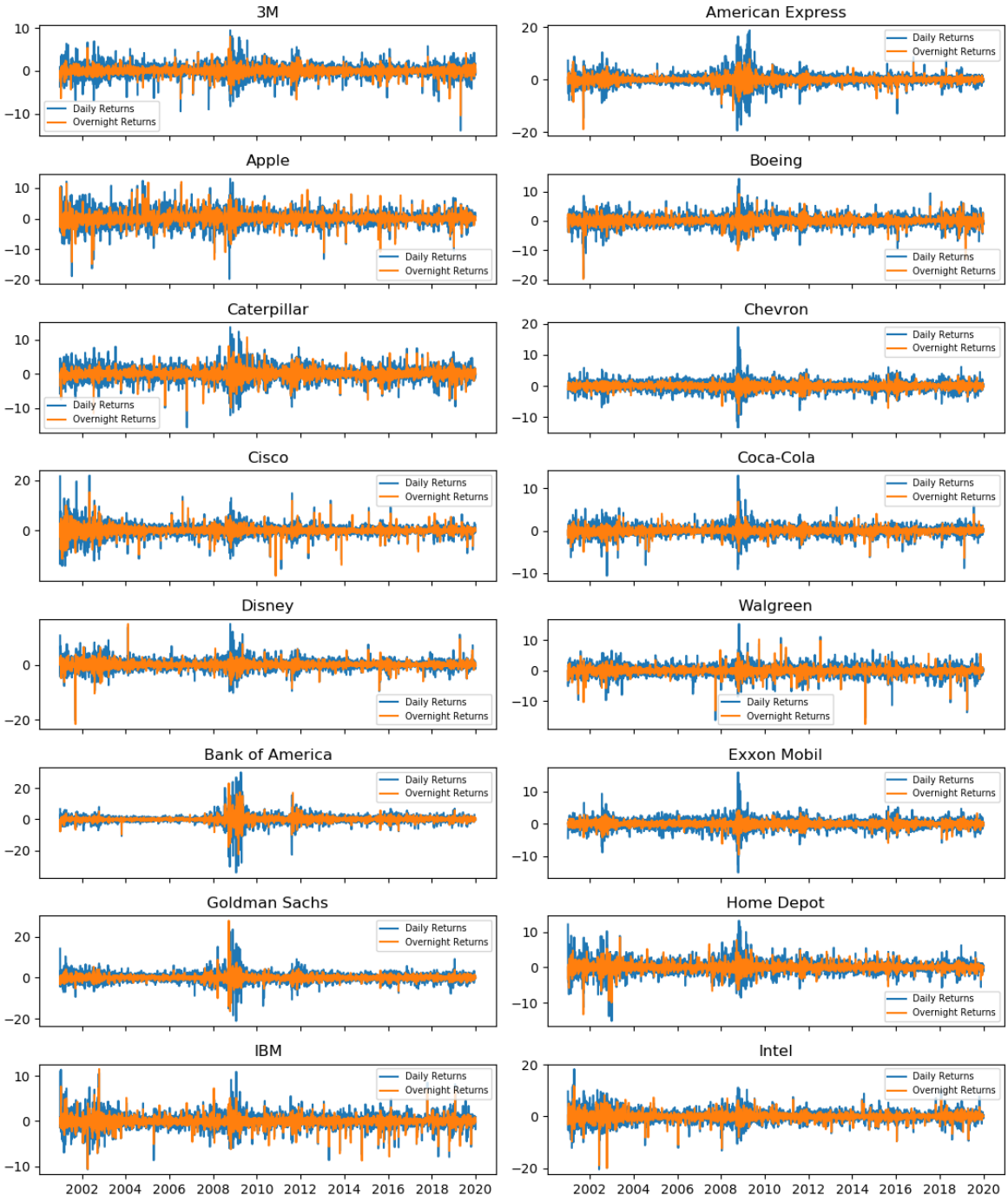


Figure 6: Overnight return plots of the DJIA.

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7.2 Summary Statistics and Plots of NASDAQ 100

	RV_t	$RV_{N,t}$	$RV_{D,t}$	BV_t	$BV_{N,t}$	$BV_{D,t}$
Mean	1.3933	0.5215	0.8718	1.3182	0.4578	0.8333
Var	11.2916	2.1037	4.5856	11.0457	1.8965	4.4340
RV_t	1.0000					
$RV_{N,t}$	0.9038	1.0000				
$RV_{D,t}$	0.9571	0.7409	1.0000			
BV_t	0.9920	0.8991	0.9477	1.0000		
$BV_{N,t}$	0.8526	0.9812	0.6732	0.8619	1.0000	
$BV_{D,t}$	0.9585	0.7552	0.9925	0.9608	0.6936	1.0000

Table 8: Summary statistics and correlations of QQQ realized volatility measures.

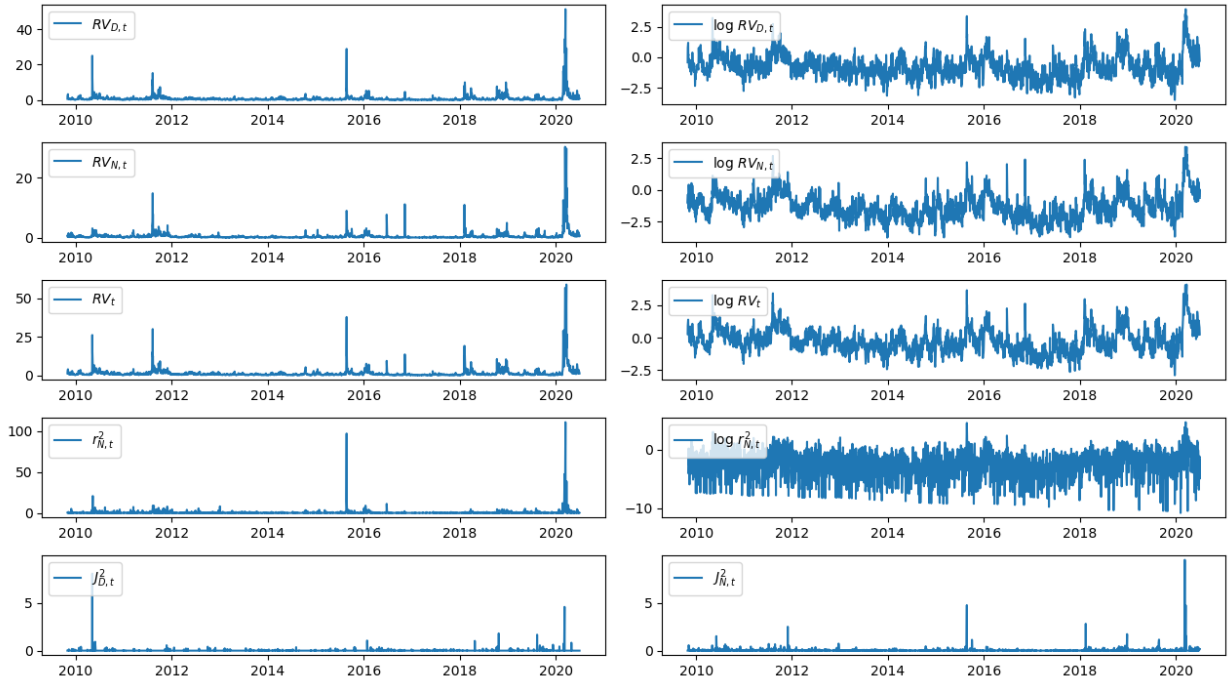


Figure 7: NADAQ 100 Realized volatility measures and squared jumps.

7.3 HAR-RV Results for NASDAQ 100

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	(1)	(2)	(3)	(4)	(5)	(6)	(7)
μ	-0.0661	-0.2120	-0.2602	-0.1031	0.0357	0.0520	-0.1455
	[0.016]	[0.024]	[0.027]	[0.021]	[0.027]	[0.030]	[0.049]
	(-0.097, -0.034)	(-0.260, -0.165)	(-0.314, -0.206)	(-0.144, -0.063)	(-0.015, 0.088)	(-0.006, 0.110)	(-0.241, -0.048)
$\log RV_{D,t-1}$	0.5647	0.4619	0.2890	0.5505	0.3304	0.3623	0.2098
	[0.018]	[0.020]	[0.023]	[0.017]	[0.020]	[0.019]	[0.022]
	(0.529, 0.600)	(0.422, 0.501)	(0.245, 0.333)	(0.516, 0.584)	(0.291, 0.370)	(0.325, 0.401)	(0.167, 0.252)
$\frac{1}{4}\sum_{i=2}^{5}\log RV_{D,t-i}$	0.2565	0.2889	0.2504	0.2463	0.2100	0.1921	0.2426
	[0.023]	[0.024]	[0.033]	[0.022]	[0.034]	[0.033]	[0.035]
	(0.211, 0.302)	(0.243, 0.335)	(0.187, 0.314)	(0.203, 0.290)	(0.143, 0.276)	(0.128, 0.256)	(0.175, 0.312)
$\frac{1}{17}\sum_{i=6}^{22}\log RV_{D,t-i}$	0.0904	0.0978	0.1387	0.0996	0.2001	0.1955	0.2269
	[0.021]	[0.021]	[0.038]	[0.020]	[0.046]	[0.045]	[0.045]
	(0.049, 0.131)	(0.057, 0.138)	(0.064, 0.214)	(0.062, 0.139)	(0.109, 0.292)	(0.107, 0.283)	(0.138, 0.315)
$r^{+}_{D,t-1}$	-	0.0622	0.0880	-	-	-	0.0749
		[0.025]	[0.024]				[0.022]
		(0.013, 0.111)	(0.041, 0.133)				(0.031, 0.117)
$r^{-}_{D,t-1}$	-	-0.2589	-0.1631	-	-	-	-0.1349
		[0.024]	[0.024]				[0.022]
		(-0.306, -0.213)	(-0.210, -0.115)				(-0.178, -0.091)
$r^{+}_{N,t}$	-	-	-	-0.1446	-	-0.1858	-0.2263
				[0.027]		[0.026]	[0.025]
				(-0.198, -0.091)		(-0.235, -0.136)	(-0.275, -0.177)
$r^{-}_{N,t}$	-	-	-	-0.2992	-	-0.1686	-0.1697
				[0.022]		[0.022]	[0.021]
				(-0.342, -0.257)		(-0.212, -0.125)	(-0.210, -0.128)
$\log RV_{N,t}$	-	-	-	-	0.4395	0.3773	0.2782
					[0.020]	[0.021]	[0.021]
					(0.400, 0.479)	(0.336, 0.418)	(0.236, 0.320)
$\frac{1}{4}\sum_{i=1}^{4}\log RV_{N,t-i}$	-	-	-	-	-0.1129	-0.0648	-0.0708
					[0.036]	[0.035]	[0.036]
					(-0.184, -0.042)	(-0.132, 0.003)	(-0.141, 0.000)
$\frac{1}{17}\sum_{i=5}^{21}\log RV_{N,t-i}$	-	-	-	-	-0.1573	-0.1457	-0.1400
					[0.044]	[0.042]	[0.049]
					(-0.245, -0.070)	(-0.228, -0.063)	(-0.236, -0.044)
$\log IV_{t-1}$	-	-	1.0921	-	-	-	0.8411
			[0.082]				[0.079]
			(0.929, 1.253)				(0.687, 0.998)
$\frac{1}{4}\sum_{i=2}^{5}\log IV_{t-i}$	-	-	-0.6243	-	-	-	-0.5349
			[0.100]				[0.095]
			(-0.816, -0.424)				(-0.720, -0.352)
$\frac{1}{17}\sum_{i=6}^{22}\log IV_{t-i}$	-	-	-0.1950	-	-	-	-0.0946
			[0.069]				[0.073]
			(-0.330, -0.061)				(-0.238, 0.047)
FMSE	0.3303	0.3163	0.2959	0.2974	0.2820	0.2652	0.2485
FMSE P.C.	-	-4.22%	-10.40%	-9.95%	-14.62%	-19.69%	-24.74%
LPL	-2253.79	-2197.06	-2106.39	-2116.39	-2048.67	-1967.98	-1882.12
LBF	-	56.73	147.41	137.40	205.12	285.81	371.67

The table reports the posterior mean, standard deviation in [] and 0.95 density interval in ().

Table 9: Forecasting NASDAQ 100 daytime logarithmic realized volatility with HAR models

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	(1)	(2)	(3)	(4)	(5)	(6)	(7)
μ	-0.0971	-0.1683	-0.3996	-0.2717	-0.1347	-0.1940	-0.4359
	[0.021]	[0.028]	[0.050]	[0.028]	[0.025]	[0.030]	[0.047]
	(-0.139, -0.055)	(-0.223, -0.115)	(-0.499, -0.301)	(-0.325, -0.217)	(-0.184, -0.086)	(-0.251, -0.135)	(-0.530, -0.342)
$\log RV_{N,t-1}$	0.4997	0.4395	0.2597	0.4519	0.2121	0.2449	0.1460
	[0.019]	[0.019]	[0.022]	[0.018]	[0.020]	[0.020]	[0.021]
	(0.463, 0.536)	(0.401, 0.478)	(0.216, 0.302)	(0.417, 0.487)	(0.174, 0.251)	(0.206, 0.284)	(0.105, 0.186)
$\frac{1}{4}\sum_{i=2}^{5}\log RV_{N,t-i}$	0.3573	0.3837	0.2763	0.3477	0.3185	0.3030	0.2783
	[0.024]	[0.024]	[0.033]	[0.022]	[0.031]	[0.030]	[0.034]
	(0.310, 0.404)	(0.337, 0.431)	(0.213, 0.342)	(0.305, 0.391)	(0.258, 0.379)	(0.243, 0.361)	(0.212, 0.345)
$\frac{1}{17}\sum_{i=6}^{22}\log RV_{N,t-i}$	0.0710	0.0806	0.1235	0.0793	0.2606	0.2549	0.2183
	[0.021]	[0.020]	[0.042]	[0.019]	[0.040]	[0.038]	[0.048]
	(0.031, 0.111)	(0.043, 0.120)	(0.039, 0.207)	(0.042, 0.116)	(0.183, 0.340)	(0.180, 0.330)	(0.124, 0.310)
$r^{+}_{N,t-1}$	-	-0.0251	-0.0542	-	-	-	0.0214
		[0.028]	[0.026]				[0.024]
		(-0.080, 0.030)	(-0.105, -0.003)				(-0.025, 0.068)
$r^{-}_{N,t-1}$	-	-0.2106	-0.1778	-	-	-	-0.1460
		[0.024]	[0.023]				[0.021]
		(-0.257, -0.164)	(-0.222, -0.133)				(-0.186, -0.105)
$r^{+}_{D,t-1}$	-	-	-	-0.0128	-	-0.0465	-0.0811
				[0.023]		[0.022]	[0.021]
				(-0.058, 0.033)		(-0.089, -0.004)	(-0.123, -0.039)
$r^{-}_{D,t-1}$	-	-	-	-0.3678	-	-0.1918	-0.1975
				[0.020]		[0.021]	[0.020]
				(-0.407, -0.328)		(-0.232, -0.151)	(-0.237, -0.158)
$\log RV_{D,t-1}$	-	-	-	-	0.4724	0.3779	0.2944
					[0.018]	[0.020]	[0.020]
					(0.438, 0.508)	(0.339, 0.416)	(0.255, 0.335)
$\frac{1}{4}\sum_{i=2}^{5}\log RV_{D,t-i}$	-	-	-	-	-0.0480	-0.0093	-0.0632
					[0.032]	[0.032]	[0.032]
					(-0.111, 0.016)	(-0.071, 0.054)	(-0.126, 0.000)
$\frac{1}{17}\sum_{i=6}^{22}\log RV_{D,t-i}$	-	-	-	-	-0.2260	-0.2129	-0.1692
					[0.043]	[0.041]	[0.041]
					(-0.312, -0.143)	(-0.295, -0.132)	(-0.249, -0.089)
$\log IV^{O}_{t-1}$	-	-	1.0257	-	-	-	0.6478
			[0.071]				[0.068]
			(0.887, 1.164)				(0.516, 0.780)
$\frac{1}{4}\sum_{i=2}^{5}\log IV^{O}_{t-i}$	-	-	-0.3847	-	-	-	-0.1773
			[0.089]				[0.082]
			(-0.556, -0.211)				(-0.338, -0.019)
$\frac{1}{17}\sum_{i=6}^{22}\log IV^{O}_{t-i}$	-	-	-0.2266	-	-	-	-0.1205
			[0.076]				[0.070]
			(-0.375, -0.078)				(-0.258, 0.015)
FMSE	0.3203	0.3098	0.2835	0.2759	0.2515	0.2400	0.2266
FMSE P.C.	-	-3.29%	-11.48%	-13.88%	-21.48%	-25.06%	-29.25%
LPL	-2212.49	-2169.60	-2053.85	-2016.84	-1897.91	-1836.92	-1763.38
LBF	-	42.89	158.64	195.65	314.57	375.57	449.11

The table reports the posterior mean, standard deviation in [] and 0.95 density interval in ().

Table 10: Forecasting NASDQA 100 overnight logarithmic realized volatility with HAR models

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